



Review

At-sea feeding ecology of parasitic lampreys

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ABSTRACT

This paper synthesizes information on the at-sea ecology of ten anadromous lampreys, with emphasis on trophic ecology. The at-sea ecology of these lampreys concerns the juvenile stage, in which growth is most rapid. Anadromous lampreys can be categorized into four groups, based on feeding modalities: 1) scavenger (Caspian lamprey, *Caspiomyzon wagneri*); 2) parasite-predator (Pacific lamprey, *Entosphenus tridentatus*); 3) predators (western river lamprey, *Lampetra ayresii*; European river lamprey, *L. fluviatilis*; Arctic lamprey, *Lethenteron camtschaticum*; pouched lamprey, *Geotria australis*; and Argentinian pouched lamprey, *G. macrostoma*); and 4) parasites (sea lamprey, *Petromyzon marinus*; Chilean lamprey, *Mordacia lapicida*; and short-headed lamprey, *M. mordax*). This paper discusses direct evidence for lamprey feeding ecology, as observed through lamprey-induced wounds on hosts and prey, and lamprey attachments on hosts and prey; and indirect evidence for feeding ecology, via analyses of fatty acids, stable isotopes, contaminants, and bioenergetics modelling. A part of the information presented on feeding ecology is from landlocked sea lamprey, and in some instances this information can be generalizable to anadromous populations. For most anadromous lampreys, but particularly for Southern Hemisphere taxa, little is known about their feeding ecology at sea. Duration of the trophic marine phase and habitat use are still subjects of debate. Species identified as lamprey hosts can be demersal or pelagic, possibly reflecting marine habitat preferences. To unlock understanding of the marine phase of anadromous lampreys, direct evidence of feeding ecology should be coupled with natural (i.e., biomarkers) and artificial (e.g., biologgers) markers to identify habitat use, movement patterns and dispersal.

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Introduction

Lampreys (Petromyzontiformes) are a small group of 42–46 species partitioned between Northern and Southern hemispheres and occurring almost exclusively in the temperate zone (Hardisty, 1986a; Docker and Potter, 2019; Mateus et al., 2013; Riva-Rossi et al., 2020; Renaud, 2011; Tutman et al., 2017). Among the 19 parasitic lamprey species, 10 are anadromous with an oceanic juvenile feeding phase; six species occur in the Northern Hemisphere (Caspian lamprey *Caspiomyzon wagneri*, Pacific lamprey *Entosphenus tridentatus*, western river lamprey *Lampetra ayresii*, European river lamprey *Lampetra fluviatilis*, Arctic lamprey *Lethenteron camtschaticum*, sea lamprey *Petromyzon marinus*) and four in the Southern Hemisphere (pouched lamprey *Geotria australis*, Chilean lamprey *Mordacia lapicida*, short-headed lamprey *Mordacia mordax*; Renaud, 2011) considering the Argentinian pouched lamprey *Geotria macrostoma* that was recently recognized with molecular and morphological evidence (Riva-Rossi et al., 2020).

The life cycle of anadromous lampreys can be divided into three distinct phases: a larval freshwater rearing, a juvenile ocean parasite and an adult freshwater spawning stage. The ecology of larval and adult lampreys during the freshwater life phases is reasonably well known (e.g., Dawson et al., 2015; Docker et al., 2015; Johnson et al., 2015; Moser et al., 2015a, 2015b). However, the trophic phase at sea is largely unknown.

Direct research on the feeding habits of lampreys in marine ecosystems is lacking (Renaud et al., 2009). This is because the gut contents of parasitic lampreys are largely unidentifiable. Host-parasite relationships have been inferred from observations of lampreys attached to hosts (e.g., Halliday, 1991; Jensen and Schwartz, 1994; Nichols and Tschertter, 2011), and lamprey wounds and scars on hosts (e.g., Beamish, 1980b; Bird et al., 1994; Halliday, 1991; Powell et al., 1999; Weitkamp et al., 2015) as well as correlations between lamprey returns and fish abundance indices (Murauskas et al., 2013) and the overlap in spatial distribution between lampreys and their putative host or prey species (e.g., Siwicke and Seitz, 2018). However, *in situ* studies of host selection dynamics and comprehensive diet estimation throughout the parasitic phase have not been done for marine lampreys.

Increasingly, laboratory methods are complementing direct observation approaches to improve our understanding of the feeding ecology of parasitic lampreys. Analytical approaches such as DNA metabarcoding (Shink et al., 2019), stable isotope analysis (e.g., Adams et al., 2008; Drevnick et al., 2006), and fatty-acid profile analysis (Happel et al., 2016a, 2016b; Lança et al., 2011, 2013)

are adding considerably to our knowledge of the trophic ecology of parasitic lampreys.

The present work arises from the need to synthesize the little and patchy information available about at-sea feeding ecology of parasitic lampreys. By using what limited information is available to infer ecological insights into lampreys at sea, it advances the recent review by Renaud and Cochran (2019) and adds innovative indirect approaches and future research needs. We start with a discussion of the feeding habits of these lampreys as a function of their feeding modality (parasite, predator or scavenger). We then give an in-depth review of direct and indirect approaches to study the feeding ecology. We discuss insights learned from studying the feeding ecology of the landlocked sea lamprey in the Great Lakes and synthesize how this information can be applied to anadromous lampreys. By doing this, we try to shed some light on the parasitic phase of anadromous lampreys, but a lot still needs to be accomplished in particular for the Southern Hemisphere species and the Northern Hemisphere Caspian lamprey for which very little is known.

Juvenile lampreys: parasites or predators?

Anadromous lampreys feed on the blood, fluids, and tissues of fishes and cetaceans as juveniles following migration to marine environments and prior to returning to freshwater systems to spawn. However, food habits vary among lamprey species as a function of the morphology of their oral disk and associated feeding modality (Potter and Hilliard, 1987; Khidir and Renaud, 2003; Renaud et al., 2009). Lampreys can therefore be grouped into four feeding modes: (1) parasites (blood feeders); (2) predators (flesh feeders); (3) parasite-predators (blood-and-flesh feeders); and (4) scavengers or carrion feeders (Table 1). The morphology of the oral disk (Fig. 1) is a general indicator of the diet, but we have to assume that diet could vary considerably within a particular feeding modality.

Parasitic (blood-feeding) lampreys include sea lamprey, Chilean lamprey, and short-head lamprey (Table 1). These lampreys have large oral disks and numerous oral fimbriae and papillae, serrated edges on the W-shaped transverse lingual lamina, parenthesis- or hook-shaped longitudinal laminae, large buccal glands, and small velar tentacles that are few in number (Fig. 1) (Khidir and Renaud, 2003; Potter and Hilliard, 1987; Renaud et al., 2009). These morphological features allow for the development of a strong suctional seal around the oral disk at the attachment site to the host to prevent loss of the consumed blood, the creation of

Table 1

Feeding modality of the nine species of anadromous lamprey and their documented host or prey species during the oceanic feeding stage. Host species not listed in Renaud and Cochran (2019) are indicated with an asterisk. Strictly freshwater species documented to be hosts were not included.

Lamprey species	Documented host or prey species	References
	Parasite (blood feeder)	
Chilean lamprey <i>Mordacia lapicida</i>	Unknown	N/A
Short-headed lamprey <i>Mordacia mordax</i>	Fishes: Black bream (<i>Acanthopagrus butcheri</i>), Yellow-eye mullet (<i>Aldrichetta forsteri</i>), Snoek (<i>Thyrstites atun</i>),	Potter et al., 1968; Renaud, 2011
Sea lamprey <i>Petromyzon marinus</i>	Fishes: Dusky shark (<i>Carcharhinus obscurus</i>), Sandbar shark (<i>Carcharhinus plumbeus</i>), Basking shark (<i>Cetorhinus maximus</i>), Greenland shark (<i>Somniosus microcephalus</i>), Tiger shark (<i>Galeocerdo cuvier</i>), Blue shark (<i>Prionace glauca</i>), Common eagle ray (<i>Myliobatis aquila</i>), Gulf sturgeon (<i>Acipenser oxyrinchus</i>)*, Atlantic sturgeon (<i>Acipenser sturio</i>)*, Blueback shad (<i>Alosa aestivalis</i>), Allis shad (<i>Alosa alosa</i>), Twaite shad (<i>Alosa fallax</i>)*, Alewife (<i>Alosa pseudoharengus</i>), American shad (<i>Alosa sapidissima</i>), Atlantic wolffish (<i>Anarhichas lupus</i>), Garfish (<i>Belone belone</i>), Bogue (<i>Boops boops</i>), Atlantic menhaden (<i>Brevoortia tyrannus</i>), Golden grey mullet (<i>Chelon aurata</i>), Thinlip grey mullet (<i>Chelon ramada</i>), Atlantic herring (<i>Clupea harengus</i>), Weakfish (<i>Cynoscion regalis</i>), European seabass (<i>Dicentrarchus labrax</i>)*, Atlantic cod (<i>Gadus morhua</i>), Haddock (<i>Melanogrammus aeglefinus</i>), European hake (<i>Merluccius merluccius</i>), Striped bass (<i>Morone saxatilis</i>), Saithe (<i>Pollachius virens</i>), Bluefish (<i>Pomatomus saltatrix</i>), Greenland halibut (<i>Reinhardtius hippoglossoides</i>), Atlantic salmon (<i>Salmo salar</i>), Sea trout (<i>Salmo trutta</i>), Brook trout (<i>Salvelinus fontinalis</i>), Beaked redbfish (<i>Sebastes mentella</i>), Common sole (<i>Solea solea</i>), Bluefin tuna (<i>Thunnus thynnus</i>), Atlantic horse mackerel (<i>Trachurus trachurus</i>), Largehead hairtail (<i>Trichiurus lepturus</i>), Red hake (<i>Urophycis chuss</i>)*, Swordfish (<i>Xiphias gladius</i>) Mammals: Minke whale (<i>Balaenoptera acutorostrata</i>), Sei whale (<i>Balaenoptera borealis</i>), Fin whale (<i>Balaenoptera physalus</i>), North Atlantic right whale (<i>Eubalaena glacialis</i>), Risso's dolphin (<i>Grampus griseus</i>), Pygmy sperm whale (<i>Kogia breviceps</i>)*, Dwarf sperm whale (<i>Kogia sima</i>)*, Humpback whale (<i>Megaptera novaeangliae</i>)*, Sowerby's beaked whale (<i>Mesoplodon bidens</i>), Killer whale (<i>Orcinus orca</i>), Harbor porpoise (<i>Phocoena phocoena</i>), Sperm whale (<i>Physeter macrocephalus</i>), Common bottlenose dolphin (<i>Tursiops truncatus</i>)*, Cuvier's beaked whale (<i>Ziphius cavirostri</i>)	Beamish (1980a), Bird et al. (1994), Gallant et al. (2006), Farmer (1980), Halliday (1991), Heyning (2003), Japha (1910), Jensen and Schwartz (1994), Jensen et al. (1998), Mansueti (1962), McAlpine (2003), Moyer et al. (2020), Miočić-Stošić et al. (2020), Nichols and Hamilton (2004), Nichols and Tschertter (2011), Potter and Beamish (1977), Powell et al. (1999), Samarra et al. (2012), Silva et al. (2013a), Silva et al. (2013b), Silva et al. (2013c), Silva et al. (2014), Van Utrecht (1959)
	Predator (flesh feeder)	
Pouched lamprey <i>Geotria australis</i>	Unknown	N/A
Argentinian pouched lamprey <i>Geotria macrostoma</i>		
Western river lamprey <i>Lampetra ayresii</i>	Fishes: American shad (<i>Alosa sapidissima</i>), Pacific herring (<i>Clupea pallasii</i>), Shiner perch (<i>Cymatogaster aggregata</i>)*, Northern anchovy (<i>Engraulis mordax mordax</i>), Surf smelt (<i>Hypomesus pretiosus</i>)*, Pink salmon (<i>Oncorhynchus gorbuscha</i>)*, Chum salmon (<i>Oncorhynchus keta</i>), Coho salmon (<i>Oncorhynchus kisutch</i>), Steelhead (<i>Oncorhynchus mykiss</i>), Sockeye salmon (<i>Oncorhynchus nerka</i>), Chinook salmon (<i>Oncorhynchus tshawytscha</i>), English sole (<i>Parophrys vetulus</i>)*, Pile surfperch (<i>Rhacochilus vacca</i>)*, Longfin smelt (<i>Spirinchus thaleichthys</i>)*	Beamish (1980b), Beamish and Neville (1995), Beamish and Youson (1987), Roos et al. (1973), Weitkamp et al. (2015)
European river lamprey <i>Lampetra fluviatilis</i>	Fishes: Shad (<i>Alosa</i> spp.)*, Atlantic herring (<i>Clupea harengus</i>), Atlantic cod (<i>Gadus morhua</i>), European smelt (<i>Osmerus eperlanus</i>)*, European flounder (<i>Platichthys flesus</i>)*, Bluefish (<i>Pomatomus saltatrix</i>)*, Sea trout (<i>Salmo trutta</i>), Atlantic mackerel (<i>Scomber scombrus</i>)*, European sprat (<i>Sprattus sprattus</i>), Atlantic salmon (<i>Salmo salar</i>), Saithe (<i>Pollachius virens</i>)	Hardisty (1986b)
Arctic lamprey <i>Lethenteron camtschaticum</i>	Fishes: Pacific sand lance (<i>Ammodytes hexapterus</i>)*, Pacific herring (<i>Clupea pallasii</i>), Saffron cod (<i>Eleginus gracilis</i>), Alaska pollock (<i>Gadus chalcogrammus</i>)*, Yellowfin sole (<i>Limanda aspera</i>)*, Capelin (<i>Mallotus villosus</i>)*, Pink salmon (<i>Oncorhynchus gorbuscha</i>), Chum salmon (<i>Oncorhynchus keta</i>), Chinook salmon (<i>Oncorhynchus tshawytscha</i>)	Maitland et al. (1984), Novomodnyy and Belyaev (2002), Pickering (1978), Tuunainen et al. (1980), Shink et al. (2019)
	Parasite-predator (blood-and-flesh feeder)	
Pacific lamprey <i>Entosphenus tridentatus</i>	Fishes: Sablefish (<i>Anoplopoma fimbria</i>), Kamchatka flounder (<i>Atheresthes evermanni</i>), Arrowtooth flounder (<i>Atheresthes stomias</i>), Roughscale sole (<i>Clidoderma asperrimum</i>), Pacific herring (<i>Clupea pallasii</i>), Petrale sole (<i>Eopsetta jordani</i>)*, Walleye pollock (<i>Gadus chalcogrammus</i>), Pacific cod (<i>Gadus macrocephalus</i>), Flathead sole (<i>Hippoglossoides elassodon</i>), Pacific halibut (<i>Hippoglossus stenolepis</i>), Bigfin eelpout (<i>Lycodes corteziensis</i>)*, Pacific hake (<i>Merluccius productus</i>), Pink salmon (<i>Oncorhynchus gorbuscha</i>), Coho salmon	Abakumov (1964), Beamish (1980b), Beamish and Neville (1995), Bond et al. (1983), Clemens et al. (2019), Clemens and Wilby (1961), Golub (2007), Murauskas et al. (2013), Nemoto (1955), Novikov (1963), Orlov et al. (2007), Orlov et al. (2008), Orlov et al. (2009), Pike (1951), Prokhorov and Grachev (1965), Renaud (2011), Siwicke and Seitz (2015), Weitkamp L., NOAA, unpublished data, Weitkamp et al. (2015), Williams and Gilhousen (1968)

Table 1 (continued)

Lamprey species	Documented host or prey species	References
	(<i>Oncorhynchus kisutch</i>), Steelhead (<i>Oncorhynchus mykiss</i>), Sockeye salmon (<i>Oncorhynchus nerka</i>), Chinook salmon (<i>Oncorhynchus tshawytscha</i>), Lincod (<i>Ophiodon elongatus</i>), Starry flounder (<i>Platichthys stellatus</i>), Atka mackerel (<i>Pleurogrammus monopterygius</i>), Greenland halibut (<i>Reinhardtius hippoglossoides</i>), Rougheye rockfish (<i>Sebastes aleutianus</i>), Pacific ocean perch (<i>Sebastes alutus</i>), Shortraker rockfish (<i>Sebastes borealis</i>), Widow rockfish (<i>Sebastes entomelas</i>), Yellowmouth rockfish (<i>Sebastes reedi</i>), Jack mackerel (<i>Trachurus symmetricus</i>) [*]	
	Mammals: Sei whale (<i>Balaenoptera borealis</i>), Fin whale (<i>Balaenoptera physalus</i>), Blue whale (<i>Balaenoptera musculus</i>) [*] , Humpback whale (<i>Megaptera novaeangliae</i>), Sperm whale (<i>Physeter macrocephalus</i>)	
	Scavenger (carrion feeder)	
Caspian lamprey <i>Caspiomyzon wagneri</i>	Unknown	Appy and Anderson (1981), Nazari et al. (2017)

N/A – not applicable.

a discrete wound through the skin and tissue of their host, and the continuous production of lamphredin (an anticoagulant) to prevent their host's blood from clotting, as well as to facilitate tissue breakdown (Khidir and Renaud, 2003; Lethbridge and Potter, 1979; Potter and Hilliard, 1987; Renaud et al., 2009). Although documentation of sea lamprey hosts is growing (Silva et al., 2014; Table 1), there is a need to obtain information on the feeding ecology of the two other blood-feeding lampreys (Chilean and short-headed lamprey; Table 1).

Predatory (flesh-feeding) lampreys include Arctic lamprey, European river lamprey, western river lamprey, pouched lamprey and Argentinian pouched lamprey (Table 1). In contrast to parasitic lampreys, predatory lampreys have a smaller oral disk with fewer oral fimbriae and papillae, an enlarged median cusp on the U-shaped transverse lingual lamina, straight or parenthesis-shaped longitudinal laminae, smaller buccal glands, and abundant, large velar tentacles (Khidir and Renaud, 2003; Potter and Hilliard, 1987; Renaud et al., 2009) (Fig. 1). Well-developed teeth on the tongue-like piston allow for the removal of flesh from their prey, and the large velar tentacles prevent these large particles from entering the pharynx (Khidir and Renaud, 2003; Potter and Hilliard, 1987; Renaud et al., 2009). The buccal glands for flesh-feeding lamprey are smaller than for blood feeders, which suggests that lamphredin is not as important for flesh feeders (Potter and Hilliard, 1987; Renaud et al., 2009). Predatory lampreys tend to target small-bodied fishes (c.f. Table 1) and ingest large pieces of muscle or even the entire fish. Observed gut contents of western river, European river, and Arctic lampreys include tissue, internal organs, eggs, fins and scales (Beamish, 1980b; Beamish and Neville, 1995; Beamish and Williams, 1976; Maitland et al., 1984; Roos et al., 1973; Shink et al., 2019). Additional research is needed to identify other prey species of flesh-feeding lampreys, particularly the pouched lampreys (Table 1), and the role and impact of these lampreys in the trophic ecology of marine systems.

Lampreys that feed on blood and flesh (parasite-predators; Pacific lamprey; Table 1) or carrion (scavengers; Caspian lamprey; Table 1) represent transitional modes of feeding and thus hold an intermediate position between blood- and flesh-feeding species (Renaud et al., 2009). For example, lampreys that feed on both blood and flesh have more oral fimbriae than flesh-feeding lampreys, and fewer oral papillae than blood-feeding lampreys. This may be an adaptive compromise between the need for maintaining a sufficient suction seal of the oral disk at the host/prey attachment site and allowing suitable site(s) on the host/prey for feeding (Khidir and Renaud, 2003). Further directed, studies should be developed to examine the feeding specificity of Caspian lamprey.

Blood-feeding lampreys typically feed on larger host species than lampreys that feed on flesh (Patrick et al., 2009; Swink, 2003; Table 1) and their attacks do not always result in the death of the prey (Renaud et al., 2009). Attacks from flesh-feeding lampreys almost inevitably result in the death of the prey but, since they normally target species that congregate in large schools, only a proportion of their numbers are attacked and therefore suffer mortality (Renaud et al., 2009). Small individuals of blood-feeding species often parasitize small fishes immediately following transformation and shift to progressively larger hosts as they reach larger body sizes (Beamish, 1980b; Halliday, 1991; Orlov et al., 2008, 2009). For Northern Hemisphere lampreys, blood feeding is ancestral to flesh feeding. As a result, it appears that the shift in lamprey evolution has been to increasingly smaller species that are flesh feeders (Gill et al., 2003; Renaud, 2011; Renaud et al., 2009). The extent to which this evolutionary shift is real (or not) requires further examination and should include more in-depth analyses of the hosts and prey of anadromous lampreys.

The opportunistic, and therefore subjective, nature of documenting observations of lamprey parasitization or predation from wounds, scars, or lamprey gut contents suggests that detection bias likely skews the hosts/prey of lampreys that are currently documented, and the extent of this bias is unknown. Some putative lamprey wounds could be confused with the wounds made by other animals such as the cookie cutter shark *Isistius brasiliensis* (Jones, 1971). Also, there is often confusion over the species of lamprey that caused a particular wound (e.g., as between Arctic lamprey and Pacific lamprey; Orlov and Baitaliuk, 2016; Siwicke and Seitz, 2015). Sources of uncertainty from gut contents of predatory and scavenging lampreys include the unknown extents of gut evacuation times and standardizing the frequencies of occurrence of particular prey in the guts with the frequencies at which these prey are encountered.

To answer the question posed by the title of this subsection, anadromous lampreys cannot be placed under all-encompassing categories of “parasites” or “predators”. Rather, the literature indicates that anadromous lampreys can be split into four feeding strategies (parasites, predators, parasite-predators, and scavengers) based on the morphology of their oral disks and their feeding behaviour. However, we should not assume that a lamprey species feeds equally on both blood and flesh just because the morphology of the oral disk infers that the species is a blood-and-flesh feeder. It could be that the species feeds primarily on blood, but also supplements its diet with some flesh or vice-versa. Additional research is needed to document other hosts/prey, particularly for those species where limited data is available (e.g., Caspian, pouched, Chilean, and short-headed lampreys; Table 1).

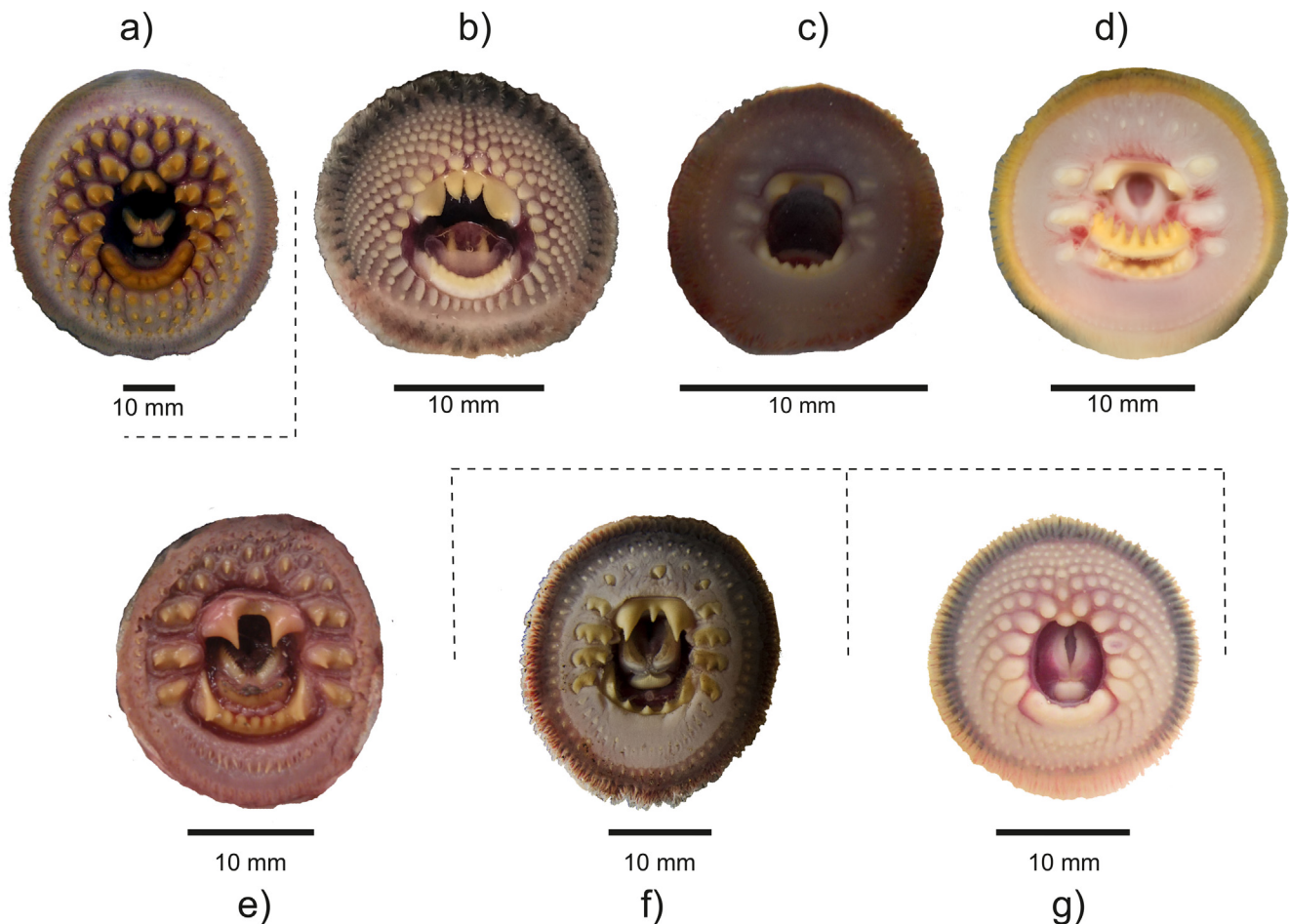


Fig. 1. The oral disks of seven anadromous lampreys, representing the typical morphology of distinct feeding modalities: blood feeder (parasite - a. Sea lamprey, *Petromyzon marinus* – Image by Bernardo Quintella); flesh feeder (predator - b. Pouched lamprey, *Geotria australis* – Image by Cindy Baker; c. western river lamprey, *Lampetra ayresii*, image by Nisga'a Fisheries & Wildlife Department; d. European river lamprey, *Lampetra fluviatilis* – Image by Bernardo Quintella; e. Arctic lamprey, *Lethenteron camtschaticum* – Image by Trent Sutton); blood-and-flesh feeder (parasite-predator - f. Pacific lamprey, *Entosphenus tridentatus* – Image by Freshwaters Illustrated via www.freshwatersillustrated.org); and carrion feeder (scavenger - g. Caspian lamprey *Caspiomyzon wagneri* – Image by Hassan Nazari).

Direct evidence of feeding ecology

Much of what we know about juvenile anadromous lampreys comes from direct evidence of their feeding ecology. For most anadromous lampreys, the available information is both sparse and scarce, and for some species, particularly the Southern Hemisphere lampreys, almost non-existent (but see Table 1). In this section, we have compiled evidence gathered by a set of approaches that, for the majority of cases, was not targeting lampreys. Within this direct evidence we include records from marine commercial fisheries, identification of species known to be hosts or prey of lampreys at sea, and identification of species known to depredate anadromous lampreys. The observations of wounds inflicted by lampreys on host species, and depredations on them can provide insights to the marine regions and habitats that lampreys occupy (Almeida and Quintella, 2013; Silva et al., 2014). The information presented in this section is organized by feeding modality and by species in an attempt to summarize what we know about the juvenile parasitic phase of each anadromous lamprey.

Blood feeders

Chilean lamprey

There is almost no information available concerning the oceanic trophic phase of Southern Hemisphere lampreys, which

includes the Chilean lamprey, the short-headed lamprey and pouched lampreys. Chilean lamprey originate in streams along the Chilean coast of southwestern South America (Renaud, 2011). The Chilean lamprey is supposedly a blood-feeding species, together with the sea lamprey and short-head lamprey (Renaud et al., 2009). Up to now, no host species have been identified (Renaud and Cochran, 2019) and there is no information on catches at sea (G. Pequeño, Universidad Austral de Chile, pers. comm.).

Short-headed lamprey

Short-headed lamprey spawn in rivers of southeastern Australia and on the island of Tasmania (Renaud, 2011). In one Australian catchment, short-headed lamprey begin their parasitic phase by feeding for the summer and autumn months in the brackish lakes and lagoons of the Gippsland Lakes system, then feed for an additional 18 months at sea before returning to fresh water to spawn (Potter et al., 1968). Up to now, there are only three marine host species described as being targeted by the juvenile parasitic lampreys (Table 1): the snoek *Thyrsites atun*, a benthopelagic marine fish usually found at depths ranging from 100 to 500 m and inhabiting continental shelves or around islands; the yellow-eye mullet *Aldrichetta forsteri* a catadromous fish that spawns in coastal waters, and the black bream *Acanthopagrus butcheri* an omnivorous estuarine sparid.

Sea lamprey

Sea lamprey originate from coastal watersheds throughout the North Atlantic Ocean basin, including enclosed seas such as the Baltic and Mediterranean seas, and also have established freshwater populations in lake watersheds of eastern North America (Renaud, 2011). More is known about the parasitic marine stage of sea lamprey than most other anadromous species (Renaud et al., 2009; Silva et al., 2014). Relatively few juvenile sea lamprey have been captured in the ocean and reports of fish with attachment scars are scarce (Farmer, 1980; Silva et al., 2014). Incidental captures of sea lamprey in fisheries are most common in the relatively shallow seas, where the majority of commercial fishing vessels operate, but this does not necessarily indicate preference for prey that live in coastal waters, given that these are fishery-dependent data that do not cover the entire bathymetric range of the potential feeding grounds (Almeida and Quintella, 2013; Lança et al., 2013). Historic transect trawling data, obtained with research vessels operating in the Celtic Sea, the North Sea and the Baltic Sea from 1977 to 2013, reported sea lamprey captures mainly from coastal waters (Heesen et al., 2015). Reported sizes ranged from 140 to 800 mm with a scarcity of individuals exceeding 400 mm. Sea lamprey were captured at depths ranging from 2 to 321 m with catch rates peaking at depths below 80 m, except in the Celtic Sea with a peak between 200 and 300 m. All individuals >400 mm were taken at depths exceeding 50 m (Heesen et al., 2015). Elliott et al. (2021) also found a relation between an increase in size of the juveniles with distance from the coast when analysing a database of 168,904 hauls (421 individuals captured predominantly <600 mm, range 130–920 mm, and observed from August to February) from both fisheries-dependent and fisheries-independent surveys between 1965 and 2019. In fishery-independent demersal surveys of the northwest Atlantic, small sea lamprey (<390 mm) were mostly captured in coastal waters or on the continental shelf, while larger individuals (>560 mm) were captured at the shelf break or on the continental slope (Halliday, 1991). These findings confirmed previous ones by Beamish (1980a), emphasizing the wide range of vertical and horizontal distribution of sea lamprey, and the tendency to encounter

small feeding individuals (<400 mm) in estuarine and coastal regions, and larger specimens in deeper water (280–650 m) off the eastern seaboard of North America. Sea lamprey have also been captured 350–400 km west of Ireland (Lelek, 1973; Kelly and King, 2001) and off the east coast of North America in samples taken from depths of 1000–4099 m (Haedrich, 1977), but we should not discount the possibility of capture at shallower depths when the net was being set or retrieved. Two juvenile sea lamprey were found feeding on blue shark *Prionace glauca* in the North Atlantic Ocean, 800 km from the nearest coast (Silva et al., 2014).

The parasitic phase of anadromous sea lamprey is generally thought to last 23–28 months (Beamish, 1980a; Halliday, 1991). However, Halliday (1991) proposed the possibility of a reduced marine feeding period for sea lamprey (1.5 years). This hypothesis was confirmed by Silva et al. (2013a) in a mark-recapture study, where a tagged sea lamprey returned to the Ulla River after an estimated feeding period of approximately 1 year (10–14 months).

Anadromous sea lamprey are associated with a wide variety of marine hosts, including bony fish, elasmobranchs and cetaceans (Beamish, 1980a; Halliday, 1991; Nichols and Hamilton, 2004; Lança et al., 2013; Silva et al., 2014), as listed in Table 1. According to the work of Silva et al. (2013b) and Silva et al. (2014), the opportunistic nature of sea lamprey feeding becomes evident when analysing the ecological, morphological and taxonomical differences among their host species. However, the large majority of the documented marine host species are oceanodromous carnivores. Documented hosts are split between demersal and pelagic neritic and pelagic oceanic species (Table 2). The main host selection factor for Great Lakes sea lamprey seems to be host size (Bence et al., 2003). Larger hosts can provide more nutritional feeding during a prolonged attachment (Farmer, 1980; Hardisty, 2006; Silva et al., 2014).

Newly transformed sea lamprey may begin parasitism in freshwater (e.g., Almeida and Quintella, 2013; Araújo et al., 2013; Davis, 1967; King and O’Gorman, 2018; Potter and Beamish, 1977), or migrate to marine waters to begin feeding and progressively move away from the river of origin as they grow, seeking larger hosts (Halliday, 1991; Hardisty, 1986c; Lança

Table 2

Classification of ecological niches of host species of lampreys, following the format of Table 2 in Silva et al. (2014). The marine or diadromous host species considered are from Table 1. The ecological information is from (Froese and Pauly, 2019) with assistance from Clemens and Wilby (1961) and Love (2011).

	Sea lamprey		Western river lamprey		European river lamprey		Arctic lamprey		Pacific lamprey	
	Fish Host Species									
Distribution range	n	%	n	%	n	%	n	%	n	%
Diadromous	12	22.64	8	57.14	5	45.45	4	44.45	6	18.75
Marine ^a	0	0	3	21.43	0	0.00	1	11.11	15	46.88
Marine (non-migratory)	3	5.66	2	14.29	0	0	3	33.33	1	3.13
Marine (oceanodromous)	38	71.70	1	7.14	6	54.54	1	11.11	10	31.25
Total	53	100.00	14	100.00	11	100.00	9	100.00	32	100.00
Habitat in the water column	n	%	n	%	n	%	n	%	n	%
Demersal	23	43.40	4	28.57	5	45.45	7	77.78	16	50.00
Pelagic ^b	0	0.00	6	42.86	0	0.00	0	0.00	10	31.25
Pelagic (neritic)	12	22.64	3	21.43	5	45.45	1	11.11	4	12.50
Pelagic (oceanic)	18	33.96	1	7.14	1	9.10	1	11.11	2	6.25
Unknown	0	0.00	0	0.00	0	0.00	0	0.00	0	0.00
Total	53	100.00	14	100.00	11	100.00	9	100.00	32	100.00
Trophic guild	n	%	n	%	n	%	n	%	n	%
Carnivore	39	73.59	11	78.57	7	63.64	6	66.67	27	84.38
Omnivore	5	9.43	2	14.29	0	0.00	0	0.00	1	3.13
Planktivore	9	16.98	1	7.14	4	36.36	3	33.33	4	12.50
Total	53	100.00	14	100.00	11	100.00	9	100.00	32	100.00

^a In the absence of clear information or information that suggests variable habitat use, this category was used instead of “Marine (non-migratory)” or “Marine (oceanodromous)”;

^b In the absence of clear information or information that suggests variable habitat use, this category was used instead of “Pelagic (neritic)” or “Pelagic (oceanic)”.

et al., 2014). According to Silva et al. (2013b), 10–30% of juvenile sea lamprey in the Ulla River of northwestern Spain begin feeding in the river before the downstream migration. The remaining 70–90% of the individuals began feeding in the estuary, particularly on the highly abundant golden grey mullet *Chelon aurata* (Silva et al., 2013b). In Iberian Peninsula sea lamprey populations, it is very common to observe juveniles feeding on *C. aurata* or *C. ramada* from November to May in estuaries and adjacent coastal areas (Almeida and Quintella, 2013; Silva et al., 2013b). According to Almeida and Quintella (2013), sea lamprey may select intermediary estuarine hosts such as mullets that, due to their migratory movements, provide a ‘ride’ (not a phoresis interaction because, besides transport, feeding occurs) with a precise seasonal timing to the continental shelf. Upon arrival at the mullets’ spawning grounds on the shelf, juvenile sea lamprey have access to a broader range of potential host species (Almeida and Quintella, 2013).

Flesh feeders

Pouched lamprey and Argentinian pouched lamprey

With the recent revalidation of the Argentinian pouched lamprey what was once considered one taxonomic entity – pouched lamprey – are now considered two distinct species (Nardi et al., 2020; Riva-Rossi et al., 2020). Pouched lamprey originate from streams in southern Australia and New Zealand while the Argentinian pouched lamprey occur in South America (Renaud, 2011). For both species, no host species have been positively identified although there are unpublished records of scars associated with *G. australis* on fresh sea-run *S. trutta* (Cindy Baker, NIWA, pers. comm.). Large numbers of Argentinian pouched lamprey (~2 million individuals during one breeding season) have been found in the regurgitated food of grey-headed albatross *Diomedea chrysostoma*, sampled in the albatross breeding colony on South Georgia Island. This suggests that this lamprey is found in large groups at sea, presumably associated with a host that is present at surface waters during the night (possibly related to a diel vertical migration of the host species) and can travel very large distances from its natal streams (Potter et al., 1979). The individuals fed on by albatrosses were large (~460 mm) and thus likely nearing the end of their parasitic phase (Potter et al., 1979). As there are no permanent rivers or streams on South Georgia or neighbouring islands, these lamprey most likely originated in rivers along the east coast of South America, approximately 1750 km from the albatross colony (Neira et al., 1988). However, albatrosses are known for their long-range foraging movements in search for food so one should not discount the possibility of the lamprey being preyed upon far away from the albatross nesting sites (Potter et al., 1979).

Western river lamprey

The western river lamprey is relatively small in body size in comparison with other anadromous lampreys. This species is distributed across a narrower geographic range compared to the Pacific lamprey – occurring in estuarine waters and associated freshwater drainages from southern California to southeastern Alaska along the west coast of North America (Renaud, 2011).

Evidence from trawl and seine catches, observations of their wounds on host species, and depredations on them suggests that western river lamprey occupies relatively shallow (<30 m depth) estuarine and marine habitats that include neritic habitats (i.e., both demersal and pelagic). At shallow depths, western river lamprey are captured in association with schools of Pacific herring (Beamish, 1980b; Wade and Beamish, 2016). Both juveniles and adults are frequently found in the ocean near large river mouths, and they may return to these rivers to spawn (Beamish, 1980b; Beamish and Youson, 1987; Kostow, 2002).

Western river lamprey feed most upon diadromous species, followed by marine species. This species feeds most upon pelagic hosts, followed by demersal hosts (Table 2). The majority of identified hosts are carnivores (Table 1).

The occurrence of western river lamprey in river mouths and estuaries likely influences their tendency to feed in schools of young Pacific salmon and Pacific herring (Beamish, 1980b; Beamish and Neville, 1995; Wade and Beamish, 2016). Juvenile Pacific salmon and Pacific herring are common prey for western river lamprey (Beamish, 1980b; Beamish and Neville, 1995) as the abundance of western river lamprey appears to be associated with the abundance of Pacific herring (Wade and Beamish, 2016). A prey switch from salmon to Pacific herring may occur following the timing and appearance of these species in the Strait of Georgia (British Columbia; Beamish, 1980b). The feeding behavior of western river lamprey (i.e., taking large chunks of flesh out of fish) suggests that they kill their prey (Beamish, 1980b; Weitkamp et al., 2015). In the Strait of Georgia, western river lamprey has been estimated to kill a significant number (11–20%) of Pacific herring, juvenile Pacific salmon and Northern anchovy, *Engraulis mordax mordax* (Beamish, 1980b; Beamish and Neville, 1995; Beamish and Youson, 1987).

European river lamprey

European river lamprey originate from many rivers in North-eastern and Western Europe, excluding Spain and Iceland (Renaud, 2011). Several studies suggest that they feed mostly in estuaries and coastal waters with lower salinity such as the Baltic Sea (Adams et al., 2008; Hardisty, 1986b; Heesen et al., 2015). Adult river lamprey were considered common along the entire southwest coast of Finland (Tuunainen et al., 1980) and in Scotland actively feeding river lamprey are considered abundant in the Forth estuary (Maitland et al., 1984). Distribution of historical catches (1649–2005) in the southern Baltic Sea indicate that offshore records of river lamprey were rare (Thiel et al., 2009). Research vessel survey trawls performed in Celtic, North and Baltic seas, reported that European river lamprey were almost exclusively caught in areas adjacent to the coast (Heesen et al., 2015). In the North Sea, most records are from the coasts of Germany and Netherlands. In the Baltic Sea, juvenile European river lamprey are frequently captured in the Kattegat area and the Gulf of Riga. Catch rates peaked in the 10–30 m depth band with captured lamprey measuring between 17 and 43 cm. Individuals caught below 20 m on average were substantially smaller than those caught above this threshold. The time-series of abundance for the North Sea region suggests that records have increased since the 1990 s, but the almost complete absence before that time raises some doubts on the proper identification (Heesen et al., 2015). According to Elliott et al. (2021), work already mentioned for the sea lamprey, the majority of the individuals were captured in autumn, within 50 km from the coast, with a mean length of 30 cm appearing to be adults returning to freshwater to spawn. Mateus et al. (2011) hypothesize that the persistence of European river lamprey in the Tagus basin (Portugal), at the southern limit of its distribution, may be due to the size of its estuary (~300 km²), which allows individuals to remain resident in the estuary to feed.

About 12 species of fishes were identified as hosts for European river lamprey (Table 1). This species fed most upon diadromous and marine oceanodromous species (Table 2). The large majority of the registered hosts have demersal and pelagic-neritic habits (Table 2). European river lamprey has a clear tendency to feed most upon carnivores, but also to a lesser extent on planktivorous fishes (Table 2).

The most common hosts for European river lamprey are clupeids and gadoids (Table 1), and the tendency to use large estuaries and coastal habitats during the juvenile stage was associated with

the presence of specific prey such as herring *Clupea harengus*, European sprat *Sprattus sprattus* and European flounder *Platichthys flesus* (Hardisty, 1986b; Sjöberg, 1980; Tuunainen et al., 1980). As flesh feeders, European river lamprey bite out pieces of host tissue rather than feeding mainly on blood (see previous section), and because of that feeding mode, guts of juveniles frequently contain fish remains, especially muscle, bone and scales of teleost fishes (Hardisty, 1986b). Maitland et al. (1984) found a positive relationship between the length of European river lamprey and size range of ingested host scales, suggesting selection of larger hosts as juveniles grew larger.

Arctic lamprey

Arctic lamprey spawn in Arctic Ocean watersheds, mainly in Russia and western North America; in many river basins of the subarctic North Pacific; and in systems as far south as Japan and the Korean Peninsula (Renaud, 2011). The current state of knowledge on Arctic lamprey food habits is largely based on visual observations of lamprey attached to fish hosts or through wounds and scars found on their hosts (Gritsenko, 1968; Heard, 1966; Nikol'skii, 1956; Novomodnyy and Belyaev, 2002; Shevlyakov and Parensky, 2010).

Arctic lamprey feed primarily on diadromous species, followed by non-migratory fish marine species. The majority of identified hosts are demersal and carnivorous (Table 2). In the Beaufort and Bering seas, Arctic lamprey have been observed to be attached to Chinook salmon, inconnu *Stenodus nelma*, starry flounder *Platichthys stellatus*, and smelts (Osmeridae; McPhail and Lindsey, 1970). Within Russian estuaries, wounds from Arctic lamprey have been detected on forage fishes (e.g., Clupeidae, Osmeridae) and juvenile Pacific salmon, which suggests the importance of these prey fishes as a food resource (Gritsenko, 1968; Nikol'skii, 1956; Novomodnyy and Belyaev, 2002; Shevlyakov and Parensky, 2010). In North America, there are observations of Arctic lamprey feeding on Pacific salmon, pygmy whitefish *Prosopium coulteri*, whitefishes *Coregonus* spp., and three-spined stickleback *Gasterosteus aculeatus* (Heard, 1966; Scott and Crossman, 1973; Renaud, 2011). However, Shink et al. (2019) found hard structures, including fin rays, vertebrae, and scales, and soft tissue during visual examinations of gut contents from marine-phase Arctic lamprey collected during trawl surveys in the eastern Bering Sea. From these gut contents, 10 fish taxa were identified using DNA metabarcoding, with the most common taxa including capelin *Mallotus villosus*, Pacific herring, Pacific sand lance *Ammodytes hexapterus*, and gadids. These results are consistent with the hypothesis that Arctic lamprey, like European and western river lampreys, is a flesh-feeding species (Renaud et al., 2009).

Although Shink et al. (2019) found evidence of Pacific salmon in the gut contents of Arctic lamprey, their importance was not as significant as noted in other studies. For example, several studies have observed lamprey wounds and scars on Pacific salmon (Beamish and Neville, 1995; Beamish and Youson, 1987; Heard, 1966; Novomodnyy and Belyaev, 2002; Shevlyakov and Parensky, 2010; Siwicke and Seitz, 2015, 2018). Based on these results, researchers have concluded that flesh-feeding lamprey species, such as Arctic lamprey, feed on Pacific salmon, and have the potential to negatively impact Pacific salmon stocks (Beamish and Neville, 1995; Beamish and Youson, 1987; Novomodnyy and Belyaev, 2002; Shevlyakov and Parensky, 2010). However, Shink et al. (2019) found that Pacific salmon were only detected in 3.5% of Arctic lamprey samples that were captured in the eastern Bering Sea. As a result, additional sampling is required to evaluate if these study results are temporally and spatially consistent.

The presence of hard structures and tissues from prey fishes in the gut contents of Arctic lamprey is consistent with flesh feeding and supports previous morphological evidence of this feeding

mode; structure of the dentition and size of the oral disk (Potter and Hilliard, 1987; Renaud et al., 2009). Although the occurrence of hard structures and internal organs in guts have been documented for other flesh-feeding lampreys (reviewed in Beamish, 1980b; Beamish and Williams, 1976; Hardisty and Potter, 1971; Maitland et al., 1984; Renaud et al., 2009), the results from Shink et al. (2019) are the first to confirm the presence of tissue masses, internal organs, and undigested hard structures in the gut contents of Arctic lamprey. The frequency in which structures were observed in the intestinal tract suggests high mortality rates for prey fishes in the eastern Bering Sea. These study results support the hypothesis that Arctic lamprey is a flesh-feeding species, even though there have been no direct observations for this species.

Blood-and-flesh feeders

Pacific lamprey

The Pacific lamprey is relatively large in body size in comparison with other anadromous lampreys. Pacific lamprey originate in North American river systems located from the southern tip of Baja California, Mexico, north to the Bering Strait, and in rivers on several Japanese islands (Renaud, 2011). At sea, Pacific lamprey occur in the North Pacific Ocean. Movements and distribution of Pacific lamprey in the ocean are not well understood (Clemens et al., 2019) as for the remaining anadromous lampreys. Clemens et al. (2019) concluded, “where they originated from or the route taken to arrive at their location, or whether observations reflect typical seasonal movements is not known. It is also unclear whether dispersal occurs primarily through independent swimming or when attached to hosts”. Evidence from trawl catches, wounds on host species, and depredations on Pacific lamprey suggests that this species occupies diverse marine ecosystems including demersal and pelagic habitats (neritic and oceanic; Clemens et al., 2019). Pacific lamprey are generally captured at shallow to mid-water depths of ~ 30–500 m, although this species has been caught down to 1485 m (Clemens et al., 2019; Orlov et al., 2008).

About 27 species of fishes and five species of mammals were identified as hosts for Pacific lamprey (Table 1). This species feeds most upon marine oceanodromous species (Table 2). Documented hosts are split between demersal and pelagic species (Table 2). Pacific lamprey feed on carnivores, omnivores, and planktivores, with a tendency to feed most upon carnivores (Table 2).

Pacific lamprey spawning run sizes in the Columbia River Basin are significantly and positively correlated with relative abundances of Pacific herring, Chinook salmon (*Oncorhynchus tshawytscha*), Pacific cod, walleye pollock, and Pacific hake (*Merluccius productus*) in the northeast Pacific Ocean (Murauskas et al., 2013). The depth of occurrence of Pacific lamprey in the ocean appears to be associated with the depth distributions of common hosts such as walleye pollock and Pacific hake (Clemens et al., 2019).

Scavengers

Caspian lamprey

The juvenile stage of the Caspian lamprey is spent in the Caspian Sea and the feeding ecology and habitat used remain largely unknown (Holčík, 1986; Nazari et al., 2017; Renaud, 2011). Renaud (1997) classified Caspian lamprey as scavengers, yet controversy remains due to the lack of ecological information during the juvenile stage as described by Holčík (1986). Up to now, there is no host species positively identified and no traceable information on catches of *C. wagneri* at sea (H. Nazari, University of South Bohemia in České Budějovice, pers. comm.). The only described attachments to potential hosts were to brown trout (*Salmo trutta*; Table 1) and Caspian trout (*Salmo caspius*) during the Caspian lamprey spawning run, but in these cases it was concluded to be for

transport only (Holčík, 1986; Nazari et al., 2017) because the attachment area was the opercular region (Renaud, 2011). The diet of juvenile Caspian lamprey is suspected to consist of benthic invertebrates, including certain amphipods, and fish eggs. Juvenile acanthocephalans (*Corynosoma* sp.) found in the gut of juvenile Caspian lamprey may suggest that they feed on amphipods, which are intermediate acanthocephalan hosts (Nazari et al., 2017). However, the feeding behavior of this species remains a controversial topic among lamprey biologists.

Indirect evidence of feeding ecology

Factors such as host survival, capture of host species by fisheries, and rapid digestion of liquid meals all contribute to difficulties in deciphering the trophic connections parasitic lamprey have with hosts and prey through observations of wounds or gut contents. As a work around, some authors have begun to use biochemical tracer or modeling methods to study trophic ecology. Such methods often rely on “You are what you eat” principles and typically involve fatty acids, stable isotopes, contaminants, or bioenergetics modeling approaches. We describe some of the current knowledge gained from these indirect methods, attempt to resolve some knowledge gaps, and isolate areas in need of further study.

Fatty acids

The ideal trophic marker is an uncommon compound that can be quantitatively extracted from biological samples, metabolically stable and easily identified with great sensitivity (Dalsgaard et al., 2003). Fatty acids (FA) have some characteristics that contribute to their role as useful tracers of diets and food-web structure: FA are released by hydrolysis from ingested lipid molecules during digestion and are taken up by tissues in their basic form and, unlike most other nutrients, stored as fat in animal tissues (Iverson, 2009). FA are conservatively transferred to the next trophic level through consumption, and hence compositional patterns of fatty acids observed in primary producers can be recognized in tissues of primary consumers (Alfaro et al., 2006; Budge et al., 2006; Dalsgaard et al., 2003). Although organisms are capable of biosynthesizing, modifying chain-length, and introducing double bonds in FA (Tocher and Glencross, 2015), these processes show limitations with increasing phylogenetic order (Cook, 1996). FA are relevant trophic markers because animals' FA compositions represent time-integrated dietary intake. For fishes, the FA profile of a predator reflects what has been consumed over the preceding two to three months but sampling different tissues may provide different time frames (Happel et al., 2016b; Kirsch et al., 1998).

The use of FA in trophic ecology has been refined in recent years (reviewed in Budge et al., 2006). Some FA are more prone to metabolism-related alterations (Strandberg et al., 2015), some fractions of lipids (i.e., neutral versus polar) respond more to dietary changes (Böhm et al., 2014; Lazzarotto et al., 2015), and different FA are preferentially stored in organ-dependent fashions (Böhm et al., 2014; McMeans et al., 2012). Further, the proportions of FA stored in a particular predator's tissue often are not identical to those of their diet, meaning some modifications occur during trophic transfer (Budge et al., 2011; Happel et al., 2019). However, such caveats have not hindered the utility of FA analysis as a technique, and several previous studies have used FA to decipher diets of wild fishes (Foley et al., 2016; Happel et al., 2015; Lau et al., 2012; Parrish et al., 2015; Pethybridge et al., 2011; Rohner et al., 2013; Strandberg et al., 2015).

Regarding lampreys, research has focused on analyzing the FA profiles of muscle, particularly the neutral lipid fraction of anadromous and landlocked sea lamprey populations. Neutral lipids

represent FA stored in tissues (e.g., triglycerides) during times of excess energy consumption and because of that they reflect the FA of diets to greater extents than other lipid classes or fractions (i.e., phospholipids or waxes; Böhm et al., 2014; Lazzarotto et al., 2015). Muscle is a commonly targeted tissue for studies because it is known as a site of high lipid storage for sea lamprey (Bird and Potter, 1983; Plisetskaya, 1980), which is a characteristic of a tissue believed to lead to a better reflection of the dietary composition. However, other lamprey tissues (e.g., eggs and liver) may also have areas of high lipid accumulation and thus their ability to provide dietary information is worth investigating.

Sea lamprey muscle FA profiles comprise large amounts of monounsaturated FA (MUFA) (Pinela et al., 2009). Some of these monounsaturated FA provide quick sources of energy for fishes that migrate long distances or during periods of starvation (Whyte et al., 1993). For example, when salmon were fed diets rich in 18:1n-9, swimming performance was shown to increase (McKenzie et al., 1998). It is possible that the high relative concentration of MUFAs in some fish species, including lampreys, represents an efficient energy store, and that these FA are used as a high-energy fuel source for long term sustained swimming. Happel et al. (2017) noted that there were not large shifts in FA profiles, or even lipid content, of muscle tissues based on short (<20 km) lamprey migrations, whereas the FA profiles of lamprey undergoing longer migrations of 25–160 km exhibited large compositional changes (Lança et al., 2011, 2013; Pinela et al., 2009). A difference is that Happel et al. (2017) analyzed total lipids instead of separating neutral or polar fractions, and the general dominance of neutral lipids (>50%, Lança et al., 2011) in lamprey muscle may have diluted any changes in the polar fraction. As such, it may be useful to separately analyze these two dominant lipid fractions (polar and neutral) to place more emphasis on trophically derived (neutral) vs structurally important (polar) FA, or to sample lamprey at the onset of migrations rather than further inland when the energy demands of migration have altered the FA profile and reduced its value for diet inference.

In Lança et al. (2013) investigated the trophic ecology of anadromous sea lamprey on the western Iberian coast using neutral lipids of sea lamprey body wall musculature. Profiles of FA revealed two compositionally different groups, suggesting two different feeding strategies within the populations studied. The sea lamprey in one group contained FA that suggested a typical pelagic food web supported by phytoplankton production, whereas the FA signatures of the other group indicated that at least part of their parasitic phase was supported by a food web with strong detrital and benthic algal pathways (Lança et al., 2013). As such it seems that lamprey that attach to a pelagic host are likely to subsequently attach to another when in the open ocean, whereas those on the continental shelf may parasitize a variety of fish ecotypes (e.g., pelagic, benthic, etc.). Similarly, FA profiles of muscle tissue in sea lamprey captured from Lake Michigan suggested a variety of prey were used as hosts (Happel et al., 2017). Happel et al. (2017) illustrated that the variability seen among Great Lakes sea lamprey FA profiles was much greater than the variation seen among FA profiles of various potential host species. These results from a landlocked population generally mirror those of Lança et al. (2013) on anadromous sea lamprey in that the species appears to utilize a variety of host species, but individuals may focus on ecotypes. These observations suggest that FA, or other biochemical tracers, could be used to investigate the potential for search image formation in lamprey and the resultant influence on habitat associations of sub-populations.

An important assumption in these studies is that blood or fluid meals reliably transfer FA profiles from host to lamprey. While little has been done to directly study how fluid meal parasites assimilate FA, diet compositions can affect FA profiles of blood (Ruyter

et al., 2000), especially when lipid-carrying cells (chylomicrons) are prevalent (Cooper et al., 2005). Some authors hypothesized that, the similarity of the FA profile of chylomicron triacylglycerols to that of the diet could be much greater than in any other tissue because only hydrolysis and esterification processes are involved in the absorption and transport of FA from diet to chylomicron triacylglycerols (Budge et al., 2016; Cook, 1996). However, there are several studies with fish that revealed that apparent digestibility varies with a FA structure (Erolđođan et al., 2013; Huguet et al., 2015; Geurden et al., 2009; Olsen et al., 1998). This means that differences in FA profile between diet and chylomicron could be found and must be considered. Thus, it is unknown to what degree blood or fluid meals transfer FA that are related to the host's identity vs the host's diet composition.

Statistical models have been constructed that allow the estimation of diet compositions using FA profiles of predators and their prey. Such models rely on mathematical corrections, termed calibration coefficients, which account for modifications to FA that occur during digestion and assimilation in predators (Galloway et al., 2015; Iverson et al., 2004). Calibration coefficients are often obtained or assessed through feeding trials (Galloway et al., 2014; Happel et al., 2016b; Nordstrom et al., 2008), although there have been estimation models developed for cases where feeding experiments are not possible (Bromaghin et al., 2017). Another challenge is that a comprehensive sampling of host species' FA profiles is required for accurate diet estimation using such models. Research is needed on the use of quantitative models on a lamprey population. Studies have instead relied on qualitative comparisons between lamprey FA profiles and those of the predator (e.g., Happel et al., 2017). A study where lampreys are reared on meals of known composition would greatly inform the utility of FA in lamprey feeding ecology.

Stable isotopes

Stable isotopes are naturally occurring variations in the neutron abundances in elements that react in predictable ways during biochemical processes (e.g., photosynthesis, cellular respiration, deamination). Trophic studies generally focus on differences among individuals or populations in standardized stable isotope ratios of carbon ($\delta^{13}\text{C}$) and nitrogen ($\delta^{15}\text{N}$), representing origins of primary production and trophic level, respectively. Carbon isotope ratios of consumer tissues are relatively similar to the source of production, with trophic enrichment (also known as fractionation or discrimination) of $\delta^{13}\text{C}$ generally -0.5 – -1‰ from source to consumer (McCutchan et al., 2003; Peterson and Howarth, 1987). This relatively small change in $\delta^{13}\text{C}$ allows discernment among possible production sources if their carbon signatures are sufficiently different, such as distinguishing freshwater, estuarine and marine sources, nearshore versus offshore sources, or pelagic versus benthic food webs (Michener and Kaufman, 2007). Nitrogen isotope ratios experience stronger trophic enrichment of 2.7 – 3.4‰ from source to consumer, making them indicators of the consumer's trophic level (Post, 2002; McCutchan et al., 2003; Vanderklift and Ponsard, 2003).

Skeletal muscle or whole-body composites are commonly used in stable isotope analyses (and are most representative of the trophic enrichment factors noted above), but tissues such as liver and blood are also used. The choice of tissues to analyze is relevant both for lampreys and hosts (e.g., Harvey et al., 2008) because lampreys may consume different tissues. Each tissue type has distinct trophic enrichment factors, also known as fractionation or discrimination. Moreover, a given tissue type has an associated time scale for data interpretation, due to differences in tissue growth and turnover rates (Michener and Kaufman, 2007). For example, skeletal muscle isotope ratios generally reflect consumption over the

past several months, whereas liver turns over faster and reflects consumption within the last month. $\delta^{13}\text{C}$ values also become more negative with greater tissue lipid content, and thus it is recommended to either apply a correction factor to the raw carbon values or extract lipids from tissues prior to $\delta^{13}\text{C}$ analysis (Post et al., 2007).

Much of the stable isotope literature on parasitic lampreys, both in freshwater and marine systems, indicates high variability in feeding ecology within and across lamprey populations. For example, Harvey et al. (2008) analyzed carbon and nitrogen isotopes and found clear spatial variation in the diets of invasive sea lamprey in Lake Superior. Inger et al. (2010) used carbon and nitrogen isotopes to show that European river lamprey in Lough Neagh, Northern Ireland rely on both native and non-native freshwater host species, but not at all upon marine-derived production. By contrast, stable isotope data revealed at least two feeding strategies of European river lamprey from River Endrick, Scotland: one consistent with feeding on freshwater hosts in adjacent Loch Lomond, and the other consistent with feeding on marine hosts (Adams et al., 2008). Anadromous adult sea lamprey returning to the Connecticut River, USA had widely ranging values of $\delta^{13}\text{C}$ (-19.3 to -16.0‰) and $\delta^{15}\text{N}$ (11.4 to 14.3‰) suggesting a diverse feeding ecology for this population (Drevnick et al., 2006). Based on stable isotope data from the food web of the adjacent Georges Bank marine ecosystem (Fry, 1988), Connecticut River sea lamprey appear to be parasitizing both pelagic fishes ($\delta^{13}\text{C} < -17.3\text{‰}$) and benthic fishes (relatively enriched $\delta^{13}\text{C} \geq -17.3\text{‰}$) (Drevnick et al., 2006).

Given data on the stable isotope ratios of a consumer, potential food sources, and appropriate trophic enrichment factors, one can estimate the proportional contributions of different food sources to consumer diet using stable isotope mixing models. The most recent mixing models use Bayesian statistics to estimate probability distributions of the proportions of potential food sources to consumer tissues (Parnell et al., 2013). These tools have been applied in many isotope studies, including some studies of larval (Evans and Bauer, 2016a, 2016b; Evans and Weber, 2020), and parasitic (Inger et al., 2010) lampreys. For parasitic European river lamprey, Inger et al. (2010) illustrated some seasonality in host selection, but a general reliance on freshwater brown trout. However, these quantitative models are sensitive to assumptions about trophic enrichment factors (Bond and Diamond, 2011) which may vary as a function of diet quality, whether the consumed tissue is solid or liquid (McCutchan et al., 2003), or as a function of consumer growth efficiency (Fry, 2006). None of these assumptions have yet been experimentally validated in lampreys, as far as we know. Estimates from these models also can have high uncertainty as the number of potential food sources increases, particularly if the number of food sources is much greater than the number of stable isotopes being analyzed (typically 2 or 3: $\delta^{15}\text{N}$, $\delta^{13}\text{C}$, and $\delta^{34}\text{S}$). Further, stable isotope information for both lampreys and host tissues is needed for these models, ideally collected during sites and periods of appropriate spatiotemporal overlap to reduce potential for sampling uncertainty and model bias. Thus, as with any model, estimates from mixing models should be evaluated with care.

Additional stable isotope sampling can benefit greatly the understanding of parasitic lamprey trophic ecology, particularly through the application of mixing models, and other mechanistic models that link stable isotope dynamics to bioenergetics, feeding and growth (Harvey et al., 2002). In addition, controlled feeding experiments would be extremely valuable for discerning the trophic enrichment factors for lampreys. Finally, an even better understanding of the relative importance of different host species may be derived from compound-specific methods, where the stable isotope ratios of specific amino acids or FA are analyzed to provide greater resolution of assimilation of different prey into

consumer tissues (Boecklen et al., 2011; Kelly and Scheibling, 2012; McMahon and McCarthy, 2016).

Contaminants

Mercury is a global pollutant with a high affinity for protein, and this contaminant accumulates in food webs (Hall et al., 1997). Mercury concentration in aquatic organisms tends to increase with increasing trophic level (Atwell et al., 1998; Vander Zanden and Rasmussen, 1996). Thus, one would expect the mercury concentration in potential hosts of sea lamprey to increase with increasing trophic level of the host.

Total mercury (Hg) determinations of adult sea lamprey caught during the spawning runs show high variability among individuals, suggesting that sources of food for juvenile sea lamprey vary substantially from one sea lamprey individual to another. For example, Drevnick et al. (2006) reported that whole-fish Hg concentration of adult sea lamprey from the Connecticut River population, which is considered to be the largest sea lamprey population in the Northwest Atlantic, ranged from 83 to 942 ng/g. These researchers partly attributed this high variation among individuals to differences in diet. This high variability in sea lamprey Hg concentrations, combined with variability in stable isotope ratios of nitrogen and carbon, led them to conclude that individuals from the Connecticut River population vary widely in their habitat use and feed on hosts from a range of trophic levels (Drevnick et al., 2006). (Pedro et al., 2014) also found high variability in concentrations of Hg in the muscle (80–2760 ng/g wet weight) of adult sea lamprey captured when entering Portuguese river basins to spawn. Like Drevnick et al. (2006), (Pedro et al., 2014) also associated this with the diversity of habitats used by the species and the multitude of potential host species selected during their parasitic marine phase. A relatively high amount of variation in muscle and whole-fish Hg concentrations has also been documented for sea lamprey from the Laurentian Great Lakes (MacEachen et al., 2000; Madenjian et al., 2014; Moses et al., 2018), suggesting that trophic level and habitat use of hosts substantially varied among sea lamprey individuals in these populations.

The relatively high Hg concentrations observed in adult sea lamprey populations support the contention that juvenile sea lamprey feed on the blood of their hosts. Blood has among the highest, or the highest, Hg concentrations of all tissues in fishes (Giblin and Massaro, 1973; Oliveira Ribeiro et al., 1999). Mean Hg concentration of adult sea lamprey from the Connecticut River population was estimated to be 465 ng/g during 2003–2004 (Drevnick et al., 2006). For the Laurentian Great Lakes adult sea lamprey, mean Hg concentration ranged from 790 ng/g in Lake Huron to 2280 ng/g in Lake Superior in 1997 (MacEachen et al., 2000), 519 ng/g in Lake Huron during 2011 (Madenjian et al., 2014), and 3010 ng/g in Lake Superior during 2013 (Moses et al., 2018).

Greater whole-fish contaminant concentrations in male adult sea lamprey compared with female adults suggest that parasitic juvenile males feed at a higher rate than parasitic juvenile females (Madenjian, 2020; Madenjian et al., 2013, 2016). Apparently, males expend energy at a faster rate than females, due to a higher resting metabolic rate (or standard metabolic rate, SMR) and greater swimming activity. This higher rate of energy expenditure induces a higher feeding rate. In turn, the higher rate of feeding in males results in a greater rate of accumulation of contaminants in males than in females for a given body size. Whole-fish polychlorinated biphenyl (PCB) and Hg concentrations in male adult sea lamprey exceeded those of females from Lake Huron by 17% and 16%, respectively, implying that juvenile males fed at a 16% greater rate than juvenile females of the same size (Madenjian et al., 2013, 2014, 2016). Higher swimming activity of male adult sea lamprey

compared with female adult sea lamprey has been documented in the laboratory (Madenjian et al., 2013).

Bioenergetics modeling

Bioenergetics models simulate metabolic and physiological processes such as feeding, respiration, growth, and reproduction. Thus, these models are useful tools in determining the energy expenditure distribution across the various components of the energy budget. Results from bioenergetics modeling indicate that juvenile sea lamprey are moderately active swimmers during the parasitic phase (Kitchell and Breck, 1980). A key component of the sea lamprey bioenergetics model developed by Kitchell and Breck (1980) is respiration, which is based on laboratory measurements by Beamish (1973), who observed little, if any, movement by sea lamprey while the sea lamprey were in the respiration chamber. Therefore, these respiration rate determinations were classified as standard metabolic rates (SMRs). Kitchell and Breck (1980) reported that the energy budget of sea lamprey from the Laurentian Great Lakes did not balance unless they assumed that the sea lamprey juveniles were physically active during the parasitic trophic phase. Hence, they multiplied SMR by an activity multiplier of 1.5 (ACT = 1.5), and in doing so they were able to balance the energy budget. Subsequent research by Cochran and Kitchell (1989) corroborated the hypothesis of Kitchell and Breck (1980). In addition, observations of sea lamprey attached to lake trout *Salvelinus namaycush* in Lake Champlain (Vermont), using an underwater camera, clearly showed that sea lamprey vigorously swim even while attached to their hosts (E. Marsden, University of Vermont, personal communication; Madenjian et al., 2013), corroborating the contention of considerable swimming activity by sea lamprey during this stage.

Bioenergetics modeling has provided additional insights into the foraging ecology of Great Lakes sea lamprey juveniles. Because the size of the individual sea lamprey juvenile relative to the size of its host appeared to be an important factor regulating mortality imposed by sea lamprey on their hosts in the Laurentian Great Lakes, individual-based models (IBMs) have been applied to these populations (MacKay, 1992; Madenjian et al., 2003). The sea lamprey bioenergetics model and a sea lamprey feeding model have been incorporated into an IBM to estimate blood consumption by the juvenile sea lamprey population and sea lamprey-induced mortality on lake trout, a favored host of sea lamprey in the Laurentian Great Lakes. Results showed that sea lamprey growth, blood consumption, and induced mortality on lake trout in Lake Huron peaked in October and November (Madenjian et al., 2003). Kitchell and Breck (1980) also concluded that sea lamprey growth and blood consumption by sea lamprey peaked in autumn, based on their bioenergetics model application to the sea lamprey population in Lake Ontario. Application of IBMs to juvenile sea lamprey populations in the Laurentian Great Lakes and Lake Champlain has also shed new light on our understanding of the lethality of sea lamprey attacks on lake trout (Madenjian et al., 2008, 2003). These IBM applications, in conjunction with observed wounding rates on lake trout and estimates of both fishing mortality and total mortality of lake trout, have suggested that lethality of a sea lamprey attack on lake trout may have been overestimated in previously published studies involving the modeling of lake trout populations to guide fisheries management (Madenjian et al., 2008).

Individual-based models and bioenergetics models may be powerful tools in tracking or predicting lamprey trophic ecology and community-level interactions in the future as global climate change results in both redistribution of species and extensions of growing seasons. For example, based on bioenergetics modeling, Cline et al. (2014) concluded that warming in Lake Superior over the past several decades had led to increases in sea lamprey size,

feeding rate and fecundity, and has likely increased host mortality rates. In addition, [Lennox et al. \(2020\)](#) used bioenergetics modeling to predict that growth of sea lamprey in Lakes Huron and Michigan would increase by 17% and 18%, respectively, by year 2070, based on projected increases in water temperature for both lakes. These predicted increases in sea lamprey weight for Lake Huron and Lake Michigan were supported by increases in blood consumption of 20% and 21%, respectively, implying that sea lamprey parasitism and host mortality rates in both lakes would be expected to increase with climate change. Such modeling exercises contribute to our understanding of the roles that parasitic sea lamprey juveniles play in aquatic and marine ecosystems as physical conditions and food webs shift in the future.

Most of this modelling work has involved applications to the landlocked sea lamprey populations in the Laurentian Great Lakes. However, these modeling applications are certainly transferable to sea lamprey populations in the Atlantic Ocean. Further, when bioenergetics models are developed for other lamprey species, these same modeling applications can be transferred to populations of these other lamprey species as well. The following two examples illustrate ways in which bioenergetics modeling may be applied to lamprey populations at-sea to better understand their feeding ecology.

Example 1: At present, insufficient data are available to develop a juvenile sea lamprey growth trajectory, with monthly or seasonal estimates of juvenile sea lamprey weight over the course of what is considered to be approximately 1–2-year residence in the Atlantic Ocean. However, if such a trajectory would become available, then, just as for the Great Lakes applications, a sea lamprey bioenergetics model can be fitted to the growth trajectory to estimate the amount of blood consumption, in monthly or seasonal increments, by an average sea lamprey while at-sea.

Example 2: Mercury poses a health threat to humans consuming sea lamprey, owing to the sea lamprey's relatively high Hg concentration ([Pedro et al., 2014](#)). By incorporating a sea lamprey bioenergetics model into a food web model for a portion of the Atlantic Ocean, the food web model would be capable of generating predic-

tions of sea lamprey Hg concentration under climate change scenarios, under scenarios of regulations on Hg emission into the atmosphere, and under various scenarios of changes in the food web. This food web model would serve as a valuable tool for fishery managers and public health makers in planning for the future. Because nearly all the Hg accumulated by sea lamprey is from dietary intake, sea lamprey Hg concentration is reflective of blood consumption rate. [Schartup et al. \(2019\)](#) used this kind of food web model approach to predict changes in Hg concentration of Atlantic bluefin tuna *Thunnus thynnus* in the North Atlantic Ocean under climate change and overfishing scenarios.

Patterns at-sea

From the little that is known, it is possible to identify some patterns in the at-sea ecology of anadromous lampreys. Smaller anadromous lampreys, such as the western river lamprey and European river lamprey, generally have more reduced distribution ranges ([Fig. 2](#)), are often associated with estuarine and coastal habitats, and appear to target diadromous host species ([Table 1](#) and [Table 2](#)). This may be related to the relatively large body sizes that diadromous fishes attain, assuming that large-bodied hosts are preferred and provide better feeding opportunity for relatively small parasitic lampreys. Additionally, at least during some periods of the life cycle, diadromous fishes concentrate in coastal waters and estuaries in relatively large numbers when entering or leaving the river basins. Conversely, larger lampreys, such as the sea lamprey, Pacific lamprey, and Arctic lamprey, with wider distribution areas ([Fig. 2](#)), seem to use a broader range of habitats and prefer oceanodromous marine hosts, but with an important component of diadromous hosts as well ([Table 1](#) and [Table 2](#)). However, comparisons of the entire distribution ranges of anadromous lampreys may over-simplify comparisons of trophic ecology if these lampreys feed primarily within a subsection of their overall distributions. For example, a variety of information seems to suggest that Pacific lamprey feed within the Bering Sea (reviewed in [Clemens et al., 2019](#)), although they can and do feed elsewhere.

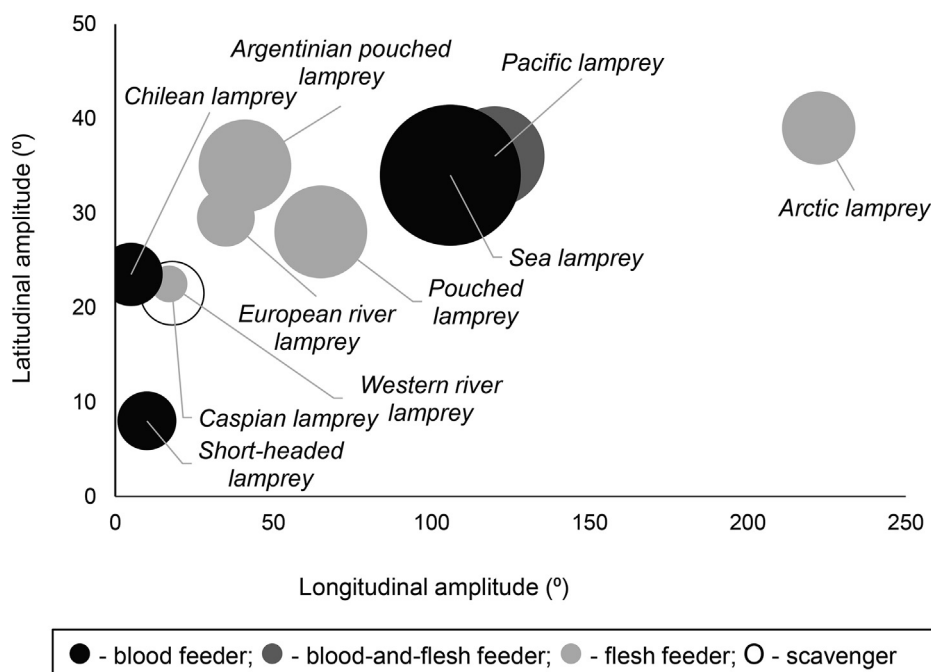


Fig. 2. Relation between the longitudinal and latitudinal distribution area of anadromous lampreys (cf [Renaud, 2011](#)). The size of the circles represents the relative maximum body length attained by each anadromous lamprey (cf [Renaud and Cochran, 2019](#); [Clemens et al., 2021](#) for Arctic lamprey). The colours of the circles represent the attributed feeding modality based on the oral disk morphology (cf [Table 1](#)).

The fate of juvenile lampreys in the ocean is linked to the choice of hosts or prey. Lampreys attached to pelagic hosts may benefit from long-distance passive dispersion to areas with productive and diverse host communities, but they may also risk not being able to return to continental waters to reproduce and complete their life cycle. A more conservative strategy, in which lampreys feed mainly on demersal hosts with limited mobility and associated with the continental slope or ocean basin floor (Lança et al., 2014), may facilitate successful return to rivers with adequate conditions for adult spawning activities and larval rearing (Almeida and Quintella, 2013). However, for the majority of the anadromous lamprey species, with the possible exception of Pacific lamprey, western river lamprey, and sea lamprey, the main host species that feed the bulk of juveniles may have not been identified yet or at least not identified as the main target species due to the lack of supporting data. Some fish families, such as Gadidae, Salmonidae, Clupeidae, Merlucciidae, Sebastidae, and Pleuronectidae, are known to be frequent hosts for several species of lampreys (Table 1). Interestingly, the majority of these families represent species with demersal habits but we should bear in mind that captures of hosts with lamprey wounds or attached lamprey are more common in shallower coastal areas, where the bulk of commercial fisheries operate (Almeida and Quintella, 2013). We must thus exercise caution when drawing conclusions about host selectivity (documented in Table 1) and host species ecological niches (Table 2) from fishery-dependent data that do not represent robust survey designs and do not cover the entire potential bathymetric range of juvenile lamprey feeding grounds (Lança et al., 2013).

The two species with the most complete record of hosts, sea lamprey and the Pacific lamprey (Table 1), are similar in that each species is captured by trawls in both benthic and pelagic environments, associated with demersal and pelagic host species respectively (Table 2; Lança et al., 2013). Lança et al. (2014) hypothesized that seabed topography during the juvenile phase induced isolation and phenotypic variation among three sea lamprey groups, as supported by differences in fatty acid profiles of heart tissues. This hypothesis is based on the assumption that the bulk of the parasitic attacks are directed towards benthic hosts, which may have restricted dispersal capability when compared with the more mobile pelagic species, and thus exchanges between lamprey feeding areas (groups) are strongly reduced (Lança et al., 2014). This limited dispersal at sea was also proposed for Pacific lamprey by Spice et al. (2012).

More research is needed to identify the feeding modes of parasitic lampreys. Controversy still remains on the scavenger type attributed to the Caspian lamprey (Renaud, 1997). The diets of anadromous lampreys could vary considerably within a particular oral disk category, meaning that flesh feeders (predators) may also feed on blood, and the converse may be true of blood feeders (parasites). Among the nine anadromous lampreys, the Pacific lamprey is the only species considered blood-and-flesh feeder (parasite-predator). No clear pattern can be observed in Fig. 2 for feeding modalities with the extension of the distribution area. Within blood feeders, sea lamprey has a wide distribution whereas the Chilean lamprey and short-headed lamprey have relatively small distribution areas. For flesh feeders, the Arctic lamprey have the larger distribution area at sea among lampreys, with three species (pouched lamprey, Argentinian pouched lamprey and European river lamprey) with a median size distribution area and the relatively smaller in size western river lamprey inhabiting a restricted area at sea. The most evident differences on the distribution area at sea are longitudinal, and this seems to be largely a function of proximity to freshwater habitats. The widest longitudinal distribution of the Arctic lamprey must be associated with the access to rivers along their distribution range spanning the Northern parts of the Asian, European and North American continents. The differ-

ences on the latitudinal axis are much smaller (Fig. 2) due to restrictions in thermal regimes and the temperate distribution of lampreys (Hardisty, 2006).

Future research

Future research needs were recently identified by Clemens et al. (2019) for the Pacific lamprey, and these are common to the remaining anadromous lampreys as well. Regarding at-sea ecology, one of the topics identified as lacking data in the Clemens et al. (2019) study is “ocean residence time”. For the majority of the anadromous lampreys, the duration of the marine phase is inferred, and the only direct knowledge of ocean residence time comes from Silva et al. (2013a) who documented an ocean residence time as short as 14 months for the sea lamprey. Short ocean residence times suggest low dispersion capacities, while longer residence times may imply a strategy that involves longer dispersion movements. Another need that Clemens et al. (2019) identified is research on “distribution, habitat use and movements”. We only know roughly the distribution area of anadromous lampreys at sea; direct evidence on habitat use comes mainly from captures at sea, which is rare (but see Orlov et al., 2008; Elliott et al., 2021). With regard to movements of juvenile lampreys at sea, scant information is available. Another topic is “population structure”, reviewed in Mateus et al. 2021, and this is particularly important for understanding the existence of distinct strategies of relatively small lamprey species with relatively small distribution areas, such as short-headed lamprey, Caspian lamprey, western river lamprey, Chilean lamprey, and European river lamprey, to larger species with broader distribution areas such as sea lamprey, Pacific lamprey and Arctic lamprey (Fig. 2). The diversity of at-sea behaviours associated with these two distinct groups of lampreys also deserves further attention. This links with another gap cited by Clemens et al. (2019), “feeding behaviour and hosts”, which largely depends on continuing to document hosts, and identifying previously undocumented host species.

We need to know much more about how lampreys encounter host species; the frequency that they change hosts; whether they have preferred hosts and engage in specific behaviours to encounter those hosts, or are merely opportunistic; and the influence of the abundance of certain host species on lamprey population size. Understanding dynamic lamprey-host interactions, combined with studies on “oceanographic limiting factors” affecting spawner abundance (Clemens et al., 2019), will be key to furthering our understanding of the at-sea ecology of lampreys. For this, specific fishery-independent research campaigns targeting the capture of these species should be conducted to avoid the sampling bias that likely arises from relying on commercial fisheries bycatch. Improved understanding of the trophic ecology of anadromous lampreys is needed across various oceanographic conditions to distinguish the influences of climate change with anthropogenic impacts (e.g., overharvest of important host and prey species; Clemens et al., 2019; Hume et al., 2020).

One potential new avenue of research that falls within the “Direct evidence of feeding ecology” involves bio-logging. Bio-loggers are considered one of the most efficient and accurate methods of assessing a species movement ecology, habitat-use and behavior (Rutz and Hays, 2009). The simplest forms of bio-loggers are archival tags that are affixed to or implanted in animals, and data are collected and stored within the tags and then manually downloaded after the deployment finishes and the archival tag is recovered (Metcalf and Arnold, 1997). Bio-loggers can collect environmental parameters (e.g., pressure, temperature) that can be used to reconstruct migration pathways and assess habitat use (Metcalf and Arnold, 1997; Heffernan et al., 2004; Hunter

et al., 2004). However, commercially available bio-loggers are still not small enough to be implanted in juvenile lampreys (<180 mm in length) at the beginning of the seaward migration to assess the vertical habitat use and dispersion at sea. Development of bio-loggers with a size and characteristics (e.g., pressure sensors) adjusted to obtain biologically useful information during the juvenile phase will considerably advance our knowledge of the at-sea ecology of lampreys. The big challenge in using bio-loggers is related to the fact that tagged lampreys will need to be recaptured when entering rivers to spawn and the devices recovered and data manually downloaded after the deployment finishes (Metcalf and Arnold, 1997). In regions where there are fisheries targeting the returning adult spawners the recapture rate may increase considerably if external tags are used.

Another worthy avenue of investigation, in this case falling within the “Indirect evidence of feeding ecology”, is the assumption that blood or fluid meals transfer biochemical information from host to parasite tissues conservatively. Research addressing this assumption would fill essential knowledge gaps in use of tracers such as stable isotopes and fatty acids in lamprey and host tissues. Feeding studies could help confirm putative tracers of certain food web pathways (e.g., benthic, pelagic, etc.), trophic enrichment values across the host – parasite transfer, and allow for evaluation of quantitative models which estimate diet compositions.

Also of interest is to further test the hypothesis proposed by Madenjian et al. (2016) that, in general, male lamprey expend energy at a higher rate than females. The higher energy expenditure rate of male lamprey suggests that they would cause more damage to their hosts than female lamprey of the same body size because the amount of injury to the host increases with increasing feeding rate of the lamprey. To determine whether SMR of males exceeds SMR of females, laboratory respirometry could be conducted in a manner similar to that of Beamish (1973). This type of testing has yet to be performed on any lamprey species. Although male adult sea lamprey have been shown to exhibit significantly higher swimming activity than female adult sea lamprey in the laboratory (Madenjian et al., 2013), a difference in swimming activity between the sexes of parasitic juveniles has yet to be directly measured. Measuring such a difference in the laboratory would be instructive, and while measuring a difference in the field would be even more instructive, determining such a difference in the field may be logistically infeasible.

In sum, we view the bio-logging of juvenile lampreys as a promising approach toward better improving our understanding the at-sea ecology of parasitic lampreys. Depending on the sensors used, this approach could potentially yield information on seasonal growth rates, resident times, movements, vertical habitat use and behavior by parasitic lampreys during their oceanic phase. Given our limited knowledge of the at-sea ecology of parasitic lampreys, this bio-logging approach, along with further advancements in indirect methods of studying at-sea ecology, should eventually provide the valuable insights needed to better manage threatened or commercially exploited lamprey species (see Clemens et al., 2021; Almeida et al., 2021).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Abakumov, V.A., 1964. Concerning the sea period of life of the Pacific lamprey *Entosphenus tridentatus* (Richardson). Trudy VNIRO 49, 253–256.
- Adams, C.E., Bissett, N., Newton, J., Maitland, P.S., 2008. Alternative migration and host parasitism strategies and their long-term stability in river lampreys from the River Endrick, Scotland. J. Fish Biol. 72, 2456–2466.
- Alfaro, A.C., Thomas, F., Sargent, L., Duxbury, M., 2006. Identification of trophic interactions within an estuarine food web (northern New Zealand) using fatty acid biomarkers and stable isotopes. Estuar. Coast. Shelf Sci. 70 (1–2), 271–286.
- Almeida, P.R., Arakawa, H., Aronsuu, K., Baker, C., Blaire, S., Beaulaton, L., Belo, A.F., Kitson, J., Kucheryavyy, A., Kynard, B., Lucas, M., Moser, M., Potaka, B., Romakkaniemi, A., Staponkus, R., Tamarapa, S., Yanai, S., Yang, G., Zhang, T., Zhuang, P., 2021. Lamprey fisheries: history, trends and management. J. Great Lakes Res. <https://doi.org/10.1016/j.jglr.2021.06.006>.
- Almeida, P.R., Quintella, B.R., 2013. Sea lamprey migration: a millennial journey. In: Ueda, H., Tsukamoto, K. (Eds.), Physiology and Ecology of Fish Migration. CRC Press, Boca Raton, FL, pp. 105–131.
- Appy, R.G., Anderson, R.C., 1981. The parasites of lampreys. In: Hardisty, M.W., Potter, I.C. (Eds.), The Biology of Lampreys. Academic Press, London, pp. 1–42.
- Araújo, M.J., Novais, D., Antunes, C., 2013. Record of a newly metamorphosed anadromous sea lamprey (*Petromyzon marinus* Linnaeus, 1758) feeding on a freshwater fish. J. Appl. Ichthyol. 29 (6), 1380–1381.
- Atwell, L., Hobson, K.A., Welch, H.E., 1998. Biomagnification and bioaccumulation of mercury in an arctic marine food web: insights from stable nitrogen isotope analysis. Can. J. Fish. Aquat. Sci. 55 (5), 1114–1121.
- Beamish, F.W.H., 1973. Oxygen consumption of adult *Petromyzon marinus* in relation to body weight and temperature. J. Fish. Board Canada 30, 1367–1370.
- Beamish, F.W.H., 1980a. Biology of the North American Anadromous sea lamprey, *Petromyzon marinus*. Can. J. Fish. Aquat. Sci. 37, 1924–1943.
- Beamish, R.J., 1980b. Adult biology of the river lamprey (*Lampetra ayresii*) and the Pacific lamprey (*Lampetra tridentata*) from the Pacific coast of Canada. Can. J. Fish. Aquat. Sci. 37, 1906–1923.
- Beamish, R.J., Williams, N.E., 1976. A preliminary report on the effects of river lamprey (*Lampetra ayresii*) predation on salmon and herring stocks. Fisheries and Marine Service Technical Report No. 611. Fisheries Research Board of Canada, Nanaimo, British Columbia.
- Beamish, R.J., Neville, C.-E., 1995. Pacific salmon and Pacific herring mortalities in the Fraser River plume caused by river lamprey (*Lampetra ayresii*). Can. J. Fish. Aquat. Sci. 52 (3), 644–650.
- Beamish, R.J., Youson, J.H., 1987. Life history and abundance of young adult *Lampetra ayresii* in the Fraser River and their possible impact on salmon and herring stocks in the Strait of Georgia. Can. J. Fish. Aquat. Sci. 44 (3), 525–537.
- Bence, J.R., Bergstedt, R.A., Christie, G.C., Cochran, P.A., Ebener, M.P., Koonce, J.F., Rutter, M.A., Swink, W.D., 2003. Sea lamprey (*Petromyzon marinus*) parasite-host interactions in the Great Lakes. J. Great Lakes Res. 29, 253–282.
- Bird, D.J., Potter, I.C., 1983. Proximate composition at various stages of adult life in the Southern Hemisphere lamprey, *Geotria australis* Gray. Comp. Biochem. Physiol. Part A Physiol. 74 (3), 623–633.
- Bird, D.J., Potter, I.C., Hardisty, M.W., Baker, B.L., 1994. Morphology, body size and behaviour of recently-metamorphosed sea lampreys, *Petromyzon marinus*, from the lower River Severn, and their relevance to the onset of parasitic feeding. J. Fish Biol. 44 (1), 67–74.
- Boecklen, W.J., Yarnes, C.T., Cook, B.A., James, A.C., 2011. On the use of stable isotopes in trophic ecology. Annu. Rev. Ecol. Evol. Syst. 42 (1), 411–440.
- Böhm, M., Schultz, S., Koussoroplis, A.-M., Kainz, M.J., Alemany, M., 2014. Tissue-specific fatty acids response to different diets in Common Carp (*Cyprinus carpio* L.). PLoS ONE 9 (4), e94759.
- Bond, A.L., Diamond, A.W., 2011. Recent Bayesian stable-isotope mixing models are highly sensitive to variation in discrimination factors. Ecol. Appl. 21 (4), 1017–1023.
- Bond, C.E., Kan, T.T., Myers, K.W., 1983. Notes on the marine life of the river lamprey, *Lampetra ayresii*, in Yaquina Bay, Oregon, and the Columbia River estuary. Fish. Bull. 81, 165–167.
- Bromaghin, J.F., Budge, S.M., Thiemann, G.W., Rode, K.D., 2017. Simultaneous estimation of diet composition and calibration coefficients with fatty acid signature data. Ecol. Evol. 7 (16), 6103–6113.
- Budge, S.M., Aucoin, L.R., Ziegler, S.E., Lall, S.P., 2016. Fractionation of stable carbon isotopes of tissue fatty acids in Atlantic pollock (*Pollachius virens*). Ecosphere 7, 1–16.

- Budge, S.M., Iverson, S.J., Koopman, H.N., 2006. Studying trophic ecology in marine ecosystems using fatty acids: a primer on analysis and interpretation. *Mar. Mammal Sci.* 22 (4), 759–801.
- Budge, S.M., Penney, S.N., Lall, S.P., 2011. Response of tissue lipids to diet variation in Atlantic salmon (*Salmo salar*): Implications for estimating diets with fatty acid analysis. *J. Exp. Mar. Biol. Ecol.* 409 (1–2), 267–274.
- Clemens, B.J., Arakawa, H., Baker, C., Coghan, S., Kucheryavyy, A., Lampman, R., Lança, M.J., Mateus, C., Nazari, H., Pequeño, G., Sutton, T., Yanai, S., 2021. Management of anadromous lampreys: common challenges, different approaches. *J. Great Lakes Res.* 47, S129–S146.
- Clemens, W.A., Wilby, G.V., 1961. Fishes of the Pacific coast of Canada. *Bull. Fish. Res. Board Can.* 68, 1–443.
- Clemens, B.J., Weitkamp, L., Wade, J., Hess, J., Harris, J., Porter, L., Siwicke, K., Parker, K., Sutton, T.M., Orlov, A.M., 2019. The marine biology of Pacific lamprey *Entosphenus tridentatus*. *Rev. Fish Biol. Fish.* 29, 767–788.
- Cline, T.J., Kitchell, J.F., Bennington, V., McKinley, G.A., Moody, E.K., Weidel, B.C., 2014. Climate impacts on landlocked sea lamprey: Implications for host-parasite interactions and invasive species management. *Ecosphere* 5 (6), art68.
- Cochran, P.A., Kitchell, J.F., 1989. A model of feeding by parasitic lampreys. *Can. J. Fish. Aquat. Sci.* 46 (11), 1845–1852.
- Cook, H.W., 1996. Fatty acid desaturation and chain elongation in eukaryotes. In: Vance, D.E., Vance, J.E. (Eds.), *Biochemistry of Lipids and Membranes*. Elsevier, Amsterdam, pp. 129–152.
- Cooper, M.H., Iverson, S.J., Heras, H., 2005. Dynamics of blood chylomicron fatty acids in a marine carnivore: implications for lipid metabolism and quantitative estimation of predator diets. *J. Comp. Physiol. B.* 175 (2), 133–145.
- Dalsgaard, J., St John, M., Kattner, G., Müller-Navarra, D., Hagen, W., 2003. Fatty acid trophic markers in the pelagic marine environment. *Adv. Mar. Biol.* 46, 225–340.
- Davis, R.M., 1967. Parasitism by newly-transformed anadromous sea lampreys on landlocked salmon and other fishes in a coastal Maine lake. *Trans. Am. Fish. Soc.* 96 (1), 11–16.
- Dawson, H.A., Quintella, B.R., Almeida, P.R., Treble, A.J., Jolley, J.C., 2015. The ecology of larval and metamorphosing lampreys. In: Docker, M.F. (Ed.), *Lampreys: Biology, Conservation and Control*. Springer, Dordrecht, pp. 75–137.
- Docker, M.F., Potter, I.C., 2019. Life history evolution in lampreys: Alternative migratory and feedings types. In: Docker, M.F. (Ed.), *Lampreys: biology, conservation and control*. vol. 2. Fish and Fisheries Series, vol. 38. Springer, Dordrecht, The Netherlands, pp. 287–410.
- Docker, M.F., Hume, J.B., Clemens, B.J., 2015. Introduction: a surfeit of lampreys. In: Docker, M.F. (Ed.), *Lampreys: Biology, Conservation and Control*. Springer Netherlands, Dordrecht, pp. 1–34.
- Drevnick, P.E., Horgan, M.J., Oris, J.T., Kynard, B.E., 2006. Ontogenetic dynamics of mercury accumulation in Northwest Atlantic sea lamprey (*Petromyzon marinus*). *Can. J. Fish. Aquat. Sci.* 63 (5), 1058–1066.
- Eroldogan, T.O., Yilmaz, A.H., Turchini, G.M., Arslan, M., Sirkecioglu, N.A., Engin, K., Özşahinoğlu, I., Mumogullarında, Pinar, 2013. Fatty acid metabolism in European sea bass (*Dicentrarchus labrax*): effects of n-6 PUFA and MUFA in fish oil replaced diets. *Fish Physiol. Biochem.* 39 (4), 941–955.
- Elliott, S.A.M., Deleys, N., Rivot, E., Acou, A., Réveillac, E., Beaulaton, L., 2021. Shedding light on the river and sea lamprey in western European marine waters. *Endang Species Res.* 44, 409–419.
- Evans, T.M., Bauer, J.E., 2016a. Using stable isotopes and C: N ratios to examine the life-history strategies and nutritional sources of larval lampreys. *J. Fish Biol.* 88 (2), 638–654.
- Evans, T.M., Bauer, J.E., 2016b. Identification of the nutritional resources of larval sea lamprey in two Great Lakes tributaries using stable isotopes. *J. Great Lakes Res.* 42 (1), 99–107.
- Evans, T.M., Weber, A.M., 2020. Multiple stable isotopes identify sediment organic matter as the primary nutritional source to American brook lamprey larvae along a stream gradient. *Can. J. Fish. Aquat. Sci.* 77 (1), 81–89.
- Farmer, G.J., 1980. Biology and physiology of feeding in adult lamprey. *Can. J. Fish. Aquat. Sci.* 37, 1751–1761.
- Foley, C., Hook, T., Happe, A., Bootsma, H., Czesny, S.J., Janssen, J., Jude, D., Rinchar, J., Henebry, M.L., 2016. Patterns of integration of invasive round (*Neogobius melanostomus*) into a nearshore food web. *Food Webs*, 10, 26–28.
- Fry, B. (Ed.), 2006. *Stable Isotope Ecology*. Springer New York, New York, NY.
- Froese, R., Pauly, D., 2019. FishBase. World Wide Web electronic publication. Accessed on the internet at <http://www.fishbase.org> on August 2019.
- Fry, B., 1988. Food web structure on Georges Bank from stable C, N, and S isotopic compositions. *Limnol. Oceanogr.* 33 (5), 1182–1190.
- Galloway, A.W.E., Brett, M.T., Holtgrieve, G.W., Ward, E.J., Ballantyne, A.P., Burns, C.W., Kainz, M.J., Müller-Navarra, D.C., Persson, J., Ravet, J.L., Strandberg, U., Taipale, S.J., Alhgren, G., Campbell, D.A., 2015. A fatty acid based bayesian approach for inferring diet in aquatic consumers. *PLoS ONE* 10 (6), e0129723.
- Galloway, A.W.E., Eisenlord, M.E., Dethier, M.N., Holtgrieve, G.W., Brett, M.T., 2014. Quantitative estimates of isopod resource utilization using a Bayesian fatty acid mixing model. *Mar. Ecol. Prog. Ser.* 507, 219–232.
- Gallant, J., Harvey-Clark, C., Myers, R.A., Stokesbury, M.J.W., 2006. Sea lamprey attached to a Greenland shark in the St. Lawrence Estuary, Canada. *Northeast. Nat.* 13 (1), 35–38.
- Geurden, I., Jutfelt, F., Olsen, R.-E., Sundell, K.S., 2009. A vegetable oil feeding history affects digestibility and intestinal fatty acid uptake in juvenile rainbow trout *Oncorhynchus mykiss*. *Comp. Biochem. Physiol. A Mol. Integr. Physiol.* 152 (4), 552–559.
- Giblin, F.J., Massaro, E.J., 1973. Pharmacodynamics of methyl mercury in the rainbow trout (*Salmo gairdneri*): tissue uptake, distribution and excretion. *Toxicol. Appl. Pharmacol.* 24 (1), 81–91.
- Gill, H.S., Renaud, C.B., Chapleau, F., Mayden, R.L., Potter, I.C., Douglas, M.E., 2003. Phylogeny of living parasitic lampreys (Petromyzontiformes) based on morphological data. *Copeia* 2003 (4), 687–703.
- Golub, E.V., 2007. Injuries of sockeye spawners from Meynypil'gyn Lake-River system, Chukotka. *Izvestiya TINRO* 149, 99–121 (In Russian).
- Gritsenko, O.F., 1968. On the question of an ecological parallelism between lampreys and salmon. *Izv. Tikhookean Nauchno-Issled. Inst. Rybn. Khoz. I. Okeanogr.* 65, 157–169 (In Russian).
- Haedrich, R.L., 1977. A sea lamprey from the deep ocean. *Copeia* 4, 767–768.
- Hall, B.D., Bodaly, R.A., Fudge, R.J.P., Rudd, J.W.M., Rosenberg, D.M., 1997. Food as the dominant pathway of methylmercury uptake by fish. *Water Air Soil Pollut.* 100, 13–24.
- Halliday, R.C., 1991. Marine distribution of the sea lamprey (*Petromyzon marinus*) in the northwest Atlantic. *Can. J. Fish. Aquat. Sci.* 48 (5), 832–842.
- Happel, A., Creque, S., Rinchar, J., Höök, T., Bootsma, H., Janssen, J., Jude, D., Czesny, S., 2015. Exploring yellow perch diets in Lake Michigan through stomach content, fatty acids, and stable isotope ratios. *J. Great Lakes Res.* 41, 172–178.
- Happel, A., Stratton, L., Kolb, C., Hays, C., Rinchar, J., Czesny, S., 2016a. Evaluating quantitative fatty acid signature analysis (QFASA) in fish using controlled feeding experiments. *Can. J. Fish. Aquat. Sci.* 73 (8), 1222–1229.
- Happel, A., Maier, C., Farese, N., Czesny, S., Rinchar, J., 2019. Fatty acids differentiate consumers despite variation within prey fatty acid profiles. *Freshw. Biol.* 64 (8), 1416–1426.
- Happel, A., Rinchar, J., Czesny, S.J., 2017. Variability in sea lamprey fatty acid profiles indicate a range of host species utilization in Lake Michigan. *J. Great Lakes Res.* 43, 182–188.
- Happel, A., Stratton, L., Patridge, R., Rinchar, J., Czesny, S., 2016b. Fatty-acid profiles of juvenile lake trout reflect experimental diets consisting of natural prey. *Freshw. Biol.* 61 (9), 1466–1476.
- Hardisty, M.W., 1986a. General introduction. In: Holčík, J. (Ed.), *The freshwater fishes of Europe, Part 1 – Petromyzontiformes*. Aula-Verlag, Wiesbaden, pp. 19–83.
- Hardisty, M.W., 1986b. *Lampetra fluviatilis* (Linnaeus, 1758). In: Holčík, J. (Ed.), *The Freshwater Fishes of Europe, Part 1 – Petromyzontiformes*. Aula-Verlag, Wiesbaden, pp. 249–278.
- Hardisty, M.W., 1986c. *Petromyzon marinus* (Linnaeus 1758). In: Holčík, J. (Ed.), *The Freshwater Fishes of Europe, Part 1 – Petromyzontiformes*. Aula-Verlag, Wiesbaden, pp. 94–116.
- Hardisty M.W. 2006. *Lampreys. life without jaws*. Forrest Text, Tresaith, UK.
- Hardisty, M.W., Potter, I.C., 1971. The general biology of adult lampreys. In: Hardisty, M.W., Potter, I.C. (Eds.), *The Biology of Lampreys*. Academic Press, London, pp. 127–206.
- Harvey, C.J., Ebner, M.P., White, C.K., 2008. Spatial and ontogenetic variability of sea lamprey diets in Lake Superior. *J. Great Lakes Res.* 34 (3), 434–449.
- Harvey, C.J., Hanson, P.C., Essington, T.E., Brown, P.B., Kitchell, J.F., 2002. Using bioenergetics models to predict stable isotope ratios in fishes. *Can. J. Fish. Aquat. Sci.* 59 (1), 115–124.
- Heard, W.R., 1966. Observations on lampreys in the Naknek River System of south west Alaska. *Copeia* 1966 (2), 332.
- Heesen, H.J.L., Daan, N., Ellis, J.R. (Eds.), 2015. *Fish Atlas of the Celtic Sea, North Sea and the Baltic Sea*. KNNV Publishing, The Netherlands.
- Heffernan, O., Righton, D., Michalsen, K., 2004. Use of data storage tags to quantify vertical movements of cod: effects on acoustic measures. *ICES J. Mar. Sci.* 61, 1062–1070.
- Heyning, J.E., 2003. Cuvier's beaked whale (*Ziphius cavirostris*). In: Wursig, W.F., Perrin, B., Theewissen, J.G.M. (Eds.), *Encyclopedia of Marine Mammal*. Academic Press, San Diego, CA, pp. 305–307.
- Holčík, J., 1986. *Caspiomyzon wagneri* (Kessler, 1870). In: Holčík, J. (Ed.), *The freshwater fishes of Europe. Petromyzontiformes*. AULA-Verlag, Wiesbaden, Germany, pp. 119–142.
- Huguet, C.T., Norambuena, F., Emery, J.A., Hermon, K., Turchini, G.M., 2015. Dietary n-6/n-3 LC-PUFA ratio, temperature and time interactions on nutrients and fatty acids digestibility in Atlantic salmon. *Aquaculture* 436, 160–166.
- Hume, J.B., Almeida, P.R., Buckley, C.M., Criger, L.A., Madenjian, C.P., Robinson, K. F., Wang, C.J., Muir, A.M., 2021. Managing native and non-native sea lamprey (*Petromyzon marinus*) through anthropogenic change: A prospective assessment of key threats and uncertainties. *J. Great Lakes Res.* 47, S704–S722.
- Hunter, E., Metcalfe, J.D., O'Brien, C.M., Arnold, G.P., Reynolds, J.D., 2004. Vertical activity patterns of free-swimming adult plaice in the southern North Sea. *Mar. Ecol. Prog. Ser.* 279, 261–273.
- Inger, R., McDonald, R.A., Rogowski, D., Jackson, A.L., Parnell, A., Preston, S.J., Harrod, C., Goodwin, C., Griffiths, D., Dick, J.T.A., Elwood, R.W., Newton, J., Bearhop, S., 2010. Do non-native invasive fish support elevated lamprey populations? *J. Appl. Ecol.* 47, 121–129.
- Iverson, S.J., 2009. Tracing aquatic food webs using fatty acids: from qualitative indicators to quantitative determination. In: Arts, M.T., Brett, M.T., Kainz, M.J. (Eds.), *Lipids in Aquatic Ecosystems*. Springer, NY, pp. 281–307.
- Iverson, S.J., Field, C., Don Bowen, W., Blanchard, W., 2004. Quantitative fatty acid signature analysis: a new method of estimating predator diets. *Ecol. Monogr.* 74 (2), 211–235.
- Japha, A., 1910. Weitere Beiträge zur Kenntnis der Walhaut (Further contributions to the knowledge of whale skin). *Zool. Jahrbücher Suppl.* 12, 711–718.

- Jensen, C., Schwartz, F.J., 1994. Atlantic Ocean occurrences of the sea lamprey, *Petromyzon marinus* (Petromyzontiformes, Petromyzontidae), parasitizing sandbar, *Carcharhinus plumbeus*, and dusky, *C. obscurus* (Carcharhiniformes: Carcharhinidae), sharks off North and South Carolina. *Brimleyana* 21, 69–72.
- Jensen, C., Schwartz, F.J., Hopkins, G., 1998. A sea lamprey (*Petromyzon marinus*)-tiger shark (*Galeocerdo cuvier*) parasitic relationship off North Carolina. *J. Elisha Mitchell Sci. Soc. Chapel Hill N. C.* 114, 72–73.
- Johnson, N.S., Buchinger, T.J., Li, W., 2015. Reproductive ecology of lampreys. In: Docker, M.F. (Ed.), *Lampreys: biology, conservation and control*. vol. 1. Fish and Fisheries Series, vol. 37. Springer, Dordrecht, The Netherlands, pp. 265–303.
- Jones, E.C., 1971. *Isistius brasiliensis*, a squaloid shark, the probable cause of crater wounds on fishes and cetaceans. *Fishery Bulletin* 69, 791–798.
- Kelly, F.L., King, J.J., 2001. A review of the ecology and distribution of three lamprey species, *L. fluviatilis*, *L. planeri* and *P. marinus*. *Biology and Environment: Proceedings of the Royal Irish Academy* 101 (3), 165–185.
- Kelly, J.R., Scheibling, R.E., 2012. Fatty acids as dietary tracers in benthic food webs. *Mar. Ecol. Prog. Ser.* 446, 1–22.
- King, J.J., O'Gorman, N., 2018. Initial observations on feeding juvenile sea lamprey (*Petromyzon marinus* L.) in Irish lakes. *Biol. Environ.* 118b, 113–120.
- Khidir, K.T., Renaud, C.B., 2003. Oral fimbriae and papillae in parasitic lampreys (Petromyzontiformes). *Envir. Biol. Fish.* 66, 271–278.
- Kirsch, P.E., Iverson, S.J., Bowen, W.D., Kerr, S.R., Ackman, R.G., 1998. Dietary effects on the fatty acid signature of whole Atlantic cod (*Gadus morhua*). *Can. J. Fish. Aquat. Sci.* 55 (6), 1378–1386.
- Kitchell, J.F., Breck, J.E., 1980. Bioenergetics model and foraging hypothesis for sea lamprey (*Petromyzon marinus*). *Can. J. Fish. Aquat. Sci.* 37 (11), 2159–2168.
- Kostow, K., 2002. In: *Oregon Lampreys: Natural History Status and Problem Analysis*. Oregon Department of Fish and Wildlife, Portland, p. 112.
- Lança, M.J., Machado, M., Ferreira, R., Alves-Pereira, I., Ruivo Quintella, B., Raposo de Almeida, P., 2013. Feeding strategy assessment through fatty acid profiles in muscles of adult sea lampreys from the western Iberian coast. *Sci. Mar.* 77 (2), 281–291.
- Lança, M.J., Machado, M., Mateus, C.S., Lourenço, M., Ferreira, A.F., Quintella, B.R., Almeida, P.R., Pond, D.W., 2014. Investigating population structure of sea lamprey (*Petromyzon marinus*, L.) in western Iberian Peninsula using morphological characters and heart fatty acid signature analyses. *PLoS ONE* 9 (9), e108110.
- Lança, M.J., Rosado, C., Machado, M., Ferreira, R., Alves-Pereira, I., Quintella, B.R., Almeida, P.R., 2011. Can muscle fatty acid signature be used to distinguish diets during the marine trophic phase of sea lamprey (*Petromyzon marinus*, L.)? *Comp. Biochem. Physiol. Part B Biochem. Mol. Biol.* 159 (1), 26–39.
- Lau, D.C.P., Vrede, T., Pickova, J., Goedkoop, W., 2012. Fatty acid composition of consumers in boreal lakes—variation across species, space and time. *Freshw. Biol.* 57, 24–38.
- Lazzarotto, V., Corraze, G., Leprevost, A., Quillet, E., Dupont-Nivet, M., Médale, F., Balcazar, J.L., 2015. Three-year breeding cycle of rainbow trout (*Oncorhynchus mykiss*) fed a plant-based diet, totally free of marine resources: consequences for reproduction, fatty acid composition and progeny survival. *PLoS ONE* 10 (2), e0117609.
- Lelek, A., 1973. Occurrence of the Sea Lamprey in Midwater off Europe. *Copeia* 1973, 136–137.
- Lennox, R.J., Bravener, G.A., Lin, H., Madenjian, C.P., Muir, A.M., Remucal, C.K., Robinson, K.F., Rous, A.M., Siefkes, M.J., Wilkie, M.P., Zielinski, D.P., Cooke, S.J., 2020. Potential changes to the biology and challenges to the management of invasive sea lamprey *Petromyzon marinus* in the Laurentian Great Lakes due to climate change. *Glob. Change Biol.* 26 (3), 1118–1137.
- Lethbridge, R.C., Potter, I.C., 1979. The oral fimbriae of the lamprey *Geotria australis*. *J. Zool.* 188, 267–277.
- Love, M.S., 2011. *Certainly more than you want to know about the Fishes of the Pacific Coast*. Really Big Press, Santa Barbara, California.
- MacEachen, C.D., Russell, R.W., Michael Whittle, D., 2000. Spatial distribution of mercury and organochlorine contaminants in Great Lakes sea lamprey (*Petromyzon marinus*). *J. Great Lakes Res.* 26 (1), 112–119.
- MacKay, N.A., 1992. Evaluating the size effects of lampreys and their hosts: application of an individual-based model. In: DeAngelis, D.L., Gross, L.J. (Eds.), *Individual Model Approaches in Ecology, Populations, Communities and Ecosystems*. CRC Press, pp. 278–294.
- Madenjian, C.P., 2020. Males exceeding females in whole-fish polychlorinated biphenyl concentration mainly attributable to higher energy expenditure rate. *Environ. Toxicol. Chem.* 39 (5), 951–952.
- Madenjian, C.P., Chipman, B.D., Marsden, J.E., 2008. New estimates of lethality of sea lamprey (*Petromyzon marinus*) attacks on lake trout (*Salvelinus namaycush*): implications for fisheries management. *Can. J. Fish. Aquat. Sci.* 65 (3), 535–542.
- Madenjian, C.P., Cochran, P.A., Bergstedt, R.A., 2003. Seasonal patterns in growth, blood consumption, and effects on hosts by parasitic-phase sea lampreys in the Great Lakes: an individual-based model approach. *J. Great Lakes Res.* 29, 332–346.
- Madenjian, C.P., Johnson, N.S., Binder, T.R., Rediske, R.R., O'Keefe, J.P., 2013. Polychlorinated biphenyl concentrations and activity of sea lamprey *Petromyzon marinus* vary by sex. *Arch. Environ. Contam. Toxicol.* 65 (4), 693–703.
- Madenjian, C.P., Johnson, N.S., Siefkes, M.J., Dettmers, J.M., Blum, J.D., Johnson, M.W., 2014. Mercury accumulation in sea lamprey (*Petromyzon marinus*) from Lake Huron. *Sci. Total Environ.* 470–471, 1313–1319.
- Madenjian, C.P., Rediske, R.R., Krabbenhoft, D.P., Stapanian, M.A., Chernyak, S.M., O'Keefe, J.P., 2016. Sex differences in contaminant concentrations of fish: a synthesis. *Biol. Sex Differ.* 7, 42.
- Maitland, P.S., Morris, K.H., East, K., Schoonoord, M.P., van der Wal, B., Potter, I.C., 1984. The estuarine biology of the River lamprey, *Lampetra fluviatilis*, in the Firth of Forth, Scotland, with particular reference to size composition and feeding. *J. Zool.* 203, 211–225.
- Mansueti, R.J., 1962. Distribution of small, newly metamorphosed sea lampreys, *Petromyzon marinus*, and their parasitism on menhaden, *Brevoortia tyrannus*, in mid-Chesapeake Bay during winter months. *Chesapeake Sci.* 3 (2), 137. <https://doi.org/10.2307/1351229>.
- Mateus, C.S., Docker, M.F., Evanno, G., Hess, J.E., Hume, J.B., Oliveira, I.C., Souissi, A., Sutton, T.M., (2021). Population structure in anadromous lampreys: Patterns and processes. *J. Great Lakes Res.* 47, S38–S58.
- Mateus, C.S., Almeida, P.R., Quintella, B.R., Alves, M.J., 2011. MtDNA markers reveal the existence of allopatric evolutionary lineages in the threatened lampreys *Lampetra fluviatilis* (L.) and *Lampetra planeri* (Bloch) in the Iberian glacial refugium. *Conserv. Genet.* 12 (4), 1061–1074.
- Mateus, C.S., Judite Alves, M., Quintella, B.R., Almeida, P.R., Vonk, R., 2013. Three new cryptic species of the lamprey genus *Lampetra* Bonnaterra, 1788 (Petromyzontiformes: Petromyzontidae) from the Iberian Peninsula. *Contrib. Zool.* 82 (1), 37–53.
- McAlpine, D., 2003. Pygmy and dwarf sperm whales (*Kogia breviceps* and *K. sima*). In: Perrin, W.F., Wursig, B., Thewissen, J.G.M. (Eds.), *Encyclopedia of Marine Mammals*. Academic Press, San Diego, CA, pp. 1007–1009.
- McCutchan, J.H., Lewis, W.M., Kendall, C., McGrath, C.C., 2003. Variation in trophic shift for stable isotope ratios of carbon, nitrogen, and sulfur. *Oikos* 102 (2), 378–390.
- McKenzie, D.J., Higgs, D.A., Dosanjh, B.S., Deacon, G., Randall, D.J., 1998. Dietary fatty acid composition influences swimming performance in Atlantic salmon (*Salmo salar*) in seawater. *Fish Physiol. Biochem.* 19, 111–122.
- McMahon, K.W., McCarthy, M.D., 2016. Embracing variability in amino acid $\delta^{15}\text{N}$ fractionation: mechanisms, implications, and applications for trophic ecology. *Ecosphere* 7 (12), e01511.
- McMeans, B.C., Arts, M.T., Fisk, A.T., 2012. Similarity between predator and prey fatty acid profiles is tissue dependent in Greenland sharks (*Somniosus microcephalus*): Implications for diet reconstruction. *J. Exp. Mar. Bio. Ecol.* 429, 55–63.
- McPhail, J.D., Lindsey, C.C., 1970. Freshwater fishes of northwestern Canada and Alaska. *Fish. Res. Board Can. Bull.* 177, 381.
- Michener, R.H., Kaufman, L., 2007. Stable isotope ratios as tracers in marine food webs: an update. *Stable Isot. Ecol. Environ. Sci.* 2, 238–282.
- Metcalfe, J.D., Arnold, G.P., 1997. Tracking fish with electronic tags. *Nature* 387 (6634), 665–666.
- Moser, M.L., Almeida, P.R., Kemp, P.S., Sorensen, P.W., 2015a. Lamprey spawning migration. In: Docker, M.F. (Ed.), *Lampreys: biology, conservation and control*. Vol. 1. Fish and Fisheries Series, Vol. 37. Springer, Dordrecht, The Netherlands, pp. 215–263.
- Moser, M.L., Jackson, A.D., Lucas, M.C., Mueller, R.P., 2015b. Behavior and potential threats to survival of migrating lamprey ammocoetes and macrophthalmia. *Rev. Fish Biol. Fish.* 25 (1), 103–116.
- Moses, S.K., Polkinghorne, C.N., Mattes, W.P., Beesley, K.M., 2018. Spatial and ontogenetic variation in mercury in Lake Superior Basin sea lamprey (*Petromyzon marinus*). *Bull. Environ. Contam. Toxicol.* 100 (1), 95–100.
- Moyer, J.K., Dodd, J., Irschick, D.J., 2020. Observation of a Sea Lamprey, *Petromyzon marinus*, on a Pelagic Blue Shark, *Prionace glauca*. *Northeastern Naturalist* 27, N16–N20.
- Murauskas, J.G., Orlov, A.M., Siwicke, K.A., 2013. Relationships between the abundance of Pacific lamprey in the Columbia River and their common hosts in the marine environment. *Trans. Am. Fish. Soc.* 142, 143–155.
- Miočić-Stošić, J., Pleslić, G., Holcer, D., 2020. Sea Lamprey (*Petromyzon marinus*) Attachment to the Common Bottlenose Dolphin (*Tursiops truncatus*). *Aquat. Mamm.* 46 (2), 152–166.
- Nardi, C.F., Sánchez, J., Fernández, D.A., Casalnuovo, M.Á., Rojo, J.H., Chalde, T., 2020. Detection of lamprey in Southernmost South America by environmental DNA (eDNA) and molecular evidence for a new species. *Polar Biol.* 43 (4), 369–383.
- Nazari, H., Abdoli, A., Kiabi, B., Renaud, C.B., 2017. Biology and conservation status of the Caspian lamprey in Iran: a review. *Bull. Lampetra, ZO ČSOP Vlašim* 8, 6–32.
- Neira, F.J., Bradley, J.S., Potter, I.C., Hilliard, R.W., 1988. Morphological variation among widely dispersed larval populations of anadromous southern hemisphere lampreys (Geotriidae and Mordaciidae). *Zool. J. Linn. Soc.* 92, 383–408.
- Nemoto, T., 1955. White scars on whales (I) lamprey marks. *Sci. Repts. Whales Res. Inst Tokyo* 10, 69–77.
- Nichols, O.C., Hamilton, P.K., 2004. Occurrence of the parasitic sea lamprey, *Petromyzon marinus*, on western North Atlantic right whales, *Eubalaena glacialis*. *Environ. Biol. Fish.* 71 (4), 413–417.
- Nichols, O.C., Tschertter, U.T., 2011. Feeding of sea lampreys *Petromyzon marinus* on minke whales *Balaenoptera acutorostrata* in the St. Lawrence Estuary, Canada. *J. Fish Biol.* 78, 338–343.
- Nikol'skii, G.B., 1956. Some data on marine life cycle of Arctic lamprey *Lampetra japonica*. *Zoologicheskii zhurnal* 34, 588–591 (In Russian).
- Nordstrom, C.A., Wilson, L.J., Iverson, S.J., Tollit, D.J., 2008. Evaluating quantitative fatty acid signature analysis (QFASA) using harbour seals *Phoca vitulina richardsi* in captive feeding studies. *Mar. Ecol. Prog. Ser.* 360, 245–263.

- Novikov, N.P., 1963. Events of attacks of Pacific lamprey *Enthosphenus tridentatus* (Gairdner) on halibuts and other fish of the Bering Sea. *Voprosy Ikhtologii* 3, 567–569.
- Novomodnyy, G.V., Belyaev, V.A. 2002. Predation by lamprey smolts *Lampetra japonica* as a main cause of Amur chum salmon and pink salmon mortality in the early sea period of life. North Pacific Anadromous Fish Commission, Technical Report 4, Vancouver.
- Oliveira Ribeiro, C.A., Rouleau, C., Pelletier, É., Audet, C., Tjälve, H., 1999. Distribution kinetics of dietary methylmercury in the arctic charr (*Salvelinus alpinus*). *Environ. Sci. Technol.* 33 (6), 902–907.
- Olsen, R.E., Henderson, R.J., Ringø, E., 1998. The digestion and selective absorption of dietary fatty acids in Arctic charr, *Salvelinus alpinus*. *Aquac. Nutr.* 4, 13–21.
- Orlov, A.M., Baitaliuk, A.A., 2016. Distribution of Arctic and Pacific lampreys in the North Pacific. In: Orlov, A., Beamish, R.J. (Eds.), *Jawless Fishes of the World*. Cambridge Scholars Publishing, Newcastle upon Tyne, pp. 32–56.
- Orlov, A.M., Vinnikov, A.V., Pelenev, D.V., 2007. Principles of studies of the sea period of anadromous parasitic lampreys (example of Pacific lamprey *Lampetra tridentata* (Gairdner, 1836) Petromyzontidae family). *Voprosy Rybolovstva* 8, 287–312.
- Orlov, A.M., Savinykh, V.F., Pelenev, D.V., 2008. Features of the spatial distribution and size structure of the Pacific lamprey *Lampetra tridentata* in the North Pacific. *Russ. J. Mar. Biol.* 34 (5), 276–287.
- Orlov, A.M., Beamish, R.J., Vinnikov, A.V., Pelenev, D., 2009. Feeding and prey of Pacific lamprey in coastal waters of the western North Pacific. In: Haro, A., Smith, K.L., Rulifson, R.A., Moffitt, C.M., Klauda, R.J., Dadswell, M.J., Cunjak, R.A., Cooper, J.E., Beal, K.L., Avery, T.S. (Eds.), *Challenges for Diadromous Fishes in a Dynamic Global Environment*. American Fisheries Society, Symposium 69, Bethesda, Maryland, pp. 875–877.
- Parnell, A.C., Phillips, D.L., Bearhop, S., Semmens, B.X., Ward, E.J., Moore, J.W., Jackson, A.L., Grey, J., Kelly, D.J., Inger, R., 2013. Bayesian stable isotope mixing models. *Environmetrics* 24, 387–399.
- Parrish, C.C., Nichols, P.D., Pethybridge, H., Young, J.W., 2015. Direct determination of fatty acids in fish tissues: quantifying top predator trophic connections. *Oecologia* 177 (1), 85–95.
- Patrick, H.K., Sutton, T.M., Swink, W.D., 2009. Host-size selection and lethality of sea lamprey on lake sturgeon. *Trans. Am. Fish. Soc.* 138, 1065–1075.
- Pedro, S., Caçador, I., Quintella, B.R., Lança, M.J., Almeida, P.R., 2014. Trace element accumulation in anadromous sea lamprey spawners. *Ecol. Freshw. Fish.* 23, 193–207.
- Peterson, B.J., Howarth, R.W., 1987. Sulfur, carbon, and nitrogen isotopes used to trace organic matter flow in the salt-marsh estuaries of Sapelo Island, Georgia 1. *Limnol. Oceanogr.* 32 (6), 1195–1213.
- Pethybridge, H., Daley, R.K., Nichols, P.D., 2011. Diet of demersal sharks and chimaeras inferred by fatty acid profiles and stomach content analysis. *J. Exp. Mar. Biol. Ecol.* 409 (1–2), 290–299.
- Pickering, A.D., 1978. Physiological aspects of the life cycle of the river lamprey, *Lampetra fluviatilis* L. In: Forty-sixth annual report for the year ended 31st March 1978. Freshwater Biological Association, Ambleside, UK, pp. 40–46.
- Pinela, S., Quintella, B.R., Almeida, P.R., Lança, M.J., 2009. Comparison of the fatty acid profile of muscle neutral lipids and phospholipids of up-river anadromous sea lamprey (*Petromyzon marinus* L.) from three Portuguese river basins. *Sci. Mar.* 73, 785–795.
- Pike, G.C., 1951. Lamprey marks on whales. *J. Fish. Board Can.* 8b (4), 275–280.
- Plisetskaya, E., 1980. Fatty acid levels in blood of cyclostomes and fish. *Environ. Biol. Fishes* 5 (3), 273–290.
- Post, D.M., 2002. Using stable isotopes to estimate trophic position: models, methods, and assumptions. *Ecology* 83 (3), 703–718.
- Post, D.M., Layman, A.A., Arrington, D.A., Takimoto, G., Quattrochi, J., Montaña, C.G., 2007. Getting to the fat of the matter: models, methods and assumptions for dealing with lipids in stable isotope analyses. *Oecologia* 152 (1), 179–189.
- Potter, I.C., Beamish, F.W.H., 1977. The freshwater biology of adult anadromous sea lampreys *Petromyzon marinus*. *J. Zool.* 181, 113–130.
- Potter, I.C., Hilliard, R.W., 1987. A proposal for the functional and phylogenetic significance of differences in the dentition of lampreys (Agnatha: Petromyzontiformes). *J. Zool.* 212, 713–737.
- Potter, I.C., Lanzing, W.J.R., Strahan, R., 1968. Morphometric and meristic studies on populations of Australian lampreys of the genus *Mordacia*. *Zool. J. Linn. Soc.* 47, 533–546.
- Potter, I.C., Prince, P.A., Croxall, J.P., 1979. Data on the adult marine and migratory phases in the life cycle of the southern hemisphere lamprey, *Geotria australis* Gray. *Environ. Biol. Fish.* 4 (1), 65–69.
- Powell, K., Trial, J.G., Dube, N., Opitz, M., 1999. External parasite infestation of sea-run Atlantic salmon (*Salmo salar*) during spawning migration in the Penobscot River, Maine, Northeast. *Nat.* 6 (4), 363. <https://doi.org/10.2307/3858276>.
- Prokhorov, V.G., Grachev, L.E., 1965. About record of Pacific lamprey *Enthosphenus tridentatus* (Gairdner) in the western Bering Sea. *Voprosy Ikhtologii* 5, 723–726 (In Russian).
- Renaud, C.B., 1997. Conservation status of Northern Hemisphere lampreys (Petromyzontidae). *J. Appl. Ichthyol.* 13, 143–148.
- Renaud, C.B., 2011. Lampreys of the World: An Annotated and Illustrated Catalogue of Lamprey Species Known to Date. Food and Agriculture Organization of the United Nations, Rome.
- Renaud, C.B., Cochran, P.A. 2019. Post-metamorphic feeding in lampreys. In: Docker, M.F. (Ed.), *Lampreys: biology, conservation and control*. vol. 2. Fish and Fisheries Series, vol. 38. Springer, Dordrecht, The Netherlands, pp. 247–285.
- Renaud, C.B., Gill, H.S., Potter, I.C., 2009. Relationships between the diets and characteristics of the dentition, buccal glands and velar tentacles of the adults of the parasitic species of lamprey. *J. Zool.* 278 (3), 231–242.
- Riva-Rossi, C., Barrasso, D.A., Baker, C., Quiroga, A.P., Baigún, C., Basso, N.G., Stephens, P.R., 2020. Revalidation of the Argentinian pouched lamprey *Geotria macrostoma* (Burmeister, 1868) with molecular and morphological evidence. *PLoS ONE* 15 (5), e0233792.
- Rohner, C.A., Couturier, L.I.E., Richardson, A.J., Pierce, S.J., Prebble, C.E.M., Gibbons, M.J., Nichols, P.D., 2013. Diet of whale sharks *Rhincodon typus* inferred from stomach content and signature fatty acid analyses. *Mar. Ecol. Prog. Ser.* 493, 219–235.
- Roos, J.F., Gilhousen, P., Killick, S.R., 1973. Parasitism on juvenile Pacific salmon (*Oncorhynchus*) and Pacific herring (*Clupea harengus pallasii*) in the Strait of Georgia by the river lamprey (*Lampetra ayresi*). *J. Fish. Res. Board Can.* 30, 565–570.
- Rutz, C., Hays, G.C., 2009. New frontiers in biologging science. *Biol. Lett.* 5 (3), 289–292.
- Ruyter, B., Røsjø, C., Einen, O., Thomassen, M.S., 2000. Essential fatty acids in Atlantic salmon: effects of increasing dietary doses of n-6 and n-3 fatty acids on growth, survival and fatty acid composition of liver, blood and carcass. *Aquac. Nutr.* 6, 119–127.
- Samarra, F.I., Fennell, A., Aoki, K., Deecke, V.B., Miller, P.J., 2012. Persistence of skin marks on killer whales (*Orcinus orca*) caused by the parasitic sea lamprey (*Petromyzon marinus*) in Iceland. *Mar. Mamm. Sci.* 28, 395–401.
- Schartup, A.T., Thackray, C.P., Qureshi, A., Dassuncao, C., Gillespie, K., Hanke, A., Sunderland, E.M., 2019. Climate change and overfishing increase neurotoxicant in marine predators. *Nature* 572 (7771), 648–650.
- Scott, W.B., Crossman, E.J., 1973. Freshwater Fishes of Canada. Fisheries Research Board of Canada, Ottawa.
- Shink, K.G., Sutton, T.M., Murphy, J.M., López, J.A., 2019. Utilizing DNA metabarcoding to characterize the diet of marine-phase Arctic lamprey (*Lethenteron camtschaticum*) in the eastern Bering Sea. *Can. J. Fish. Aquat. Sci.* 76 (11), 1993–2002.
- Shevlyakov, V.A., Porensky, V.A., 2010. Traumatization of Kamchatka River Pacific salmon by lampreys. *Russ. J. Mar. Biol.* 36 (5), 396–400.
- Silva, S., Araújo, M.J., Bao, M., Mucientes, G., Cobo, F., 2014. The haematophagous feeding stage of anadromous populations of sea lamprey *Petromyzon marinus*: low host selectivity and wide range of habitats. *Hydrobiologia* 734 (1), 187–199.
- Silva, S., Servia, M.J., Vieira-Lanero, R., Barca, S., Cobo, F., 2013a. Life cycle of the sea lamprey *Petromyzon marinus*: duration of and growth in the marine life stage. *Aquat. Biol.* 18 (1), 59–62.
- Silva, S., Servia, M.J., Vieira-Lanero, R., Cobo, F., 2013b. Downstream migration and hematophagous feeding of newly metamorphosed sea lampreys (*Petromyzon marinus* Linnaeus, 1758). *Hydrobiologia* 700 (1), 277–286.
- Silva, S., Servia, M.J., Vieira-Lanero, R., Nachón, D.J., Cobo, F., 2013c. Haematophagous feeding of newly metamorphosed European sea lampreys *Petromyzon marinus* on strictly freshwater species. *J. Fish Biol.* 82, 1739–1745.
- Siwicke, K.A., Seitz, A.C., 2015. Interpreting lamprey attacks on Pacific cod in the eastern Bering Sea. *Trans. Am. Fish. Soc.* 144, 1249–1262.
- Siwicke, K.A., Seitz, A.C., 2018. Spatial differences in the distributions of Arctic and Pacific lampreys in the eastern Bering Sea. *Trans. Am. Fish. Soc.* 147, 111–127.
- Sjöberg, K., 1980. Ecology of the European river lamprey (*Lampetra fluviatilis*) in northern Sweden. *Can. J. Fish. Aquat. Sci.* 37, 1974–1980.
- Spice, E.K., Goodman, D.H., Reid, S.B., Docker, M.F., 2012. Neither philopatric nor panmictic: microsatellite and mtDNA evidence suggests lack of natal homing but limits to dispersal in Pacific lamprey. *Mol. Ecol.* 21, 2916–2930.
- Strandberg, U., Hiltunen, M., Jekkanen, E., Taipale, S.J., Kainz, M.J., Brett, M.T., Kankaala, P., 2015. Selective transfer of polyunsaturated fatty acids from phytoplankton to planktivorous fish in large boreal lakes. *Sci. Total Environ.* 536, 858–865.
- Swink, W.D., 2003. Host selection and lethality of attacks by sea lampreys (*Petromyzon marinus*) in laboratory studies. *J. Great Lakes Res.* 29, 307–319.
- Thiel, R., Winkler, H.M., Riel, P., Neumann, R., Gröhsler, T., Böttcher, U., Spratte, S., Hartmann, U., 2009. Endangered anadromous lampreys in the southern Baltic Sea: Spatial distribution, long-term trend, population status. *Endanger. Species Res.* 8, 233–247.
- Tocher, D.R., Glencross, B.D., 2015. Lipids and Fatty Acids. In: Lee, C.-S., Lim, C., Gatlin, D.M., Webster, C. (Eds.), *Dietary Nutrients, Additives and Fish Health*. John Wiley & Sons Inc, Hoboken, NJ, USA.
- Tutman, P., Freyhof, J., Dulcic, J., Glamuzina, B., Geiger, M., 2017. *Lampetra soljani*, a new brook lamprey from the southern Adriatic Sea basin (Petromyzontiformes: Petromyzontidae). *Zootaxa* 4273, 531–548.
- Tuunainen, P., Ikonen, E., Auvinen, H., 1980. Lampreys and lamprey fisheries in Finland. *Can. J. Fish. Aquat. Sci.* 37, 1953–1959.
- van Utrecht, W.L., 1959. Wounds and scars in the skin of the common porpoise, *Phocoena phocaena* (L.). *Mammalia* 13, 100–122.
- Vander Zanden, M.J., Rasmussen, J.B., 1996. A trophic position model of pelagic food webs: impact on contaminant bioaccumulation in lake trout. *Ecol. Monogr.* 66, 451–477.
- Vanderklift, M.A., Ponsard, S., 2003. Sources of variation in consumer-diet $\delta^{15}\text{N}$ enrichment: a meta-analysis. *Oecologia* 136, 169–182.

- Wade, J., Beamish, R., 2016. Trends in the catches of River and Pacific lamprey in the Strait of Georgia. In: Orlov, A.M., Beamish, R.J. (Eds.), *Jawless Fishes of the World*. Cambridge Scholars Publishing, Newcastle upon Tyne, pp. 57–72.
- Weitkamp, L.A., Hinton, S.A., Bentley, P.J., 2015. Seasonal abundance, size, and host selection of Western river (*Lampetra ayresii*) and Pacific (*Entosphenus tridentatus*) lampreys in the Columbia River estuary. *Fish. Bull.* 113, 213–226.
- Whyte, J.N.C., Beamish, R.J., Ginther, N.G., Neville, C.-E., 1993. Nutritional Condition of the Pacific Lamprey (*Lampetra tridentata*) Deprived of Food for Periods of Up to Two Years. *Can. J. Fish. Aquat. Sci.* 50, 591–599.
- Williams, I.V., Gilhousen, P., 1968. Lamprey parasitism on Fraser River sockeye and pink salmon during 1967. In: *International Pacific Salmon Fisheries Commission Progress Report*, p. 18.